# Chapter 5 : Elements of Functional Analysis

Arkaraj Mukherjee

July 13, 2026

# Contents

**Chapter 5 : Elements of Functional Analysis** **4**
Exercise 5.1 . . . 4
Exercise 5.2 . . . 4
Exercise 5.3 . . . 4
Exercise 5.4 . . . 5
Exercise 5.5 . . . 5
Exercise 5.6 . . . 5
Exercise 5.7 . . . 6
Exercise 5.8 . . . 6
Exercise 5.9 . . . 7
Exercise 5.10 . . . 7
Exercise 5.11 . . . 8
Exercise 5.12 . . . 9
Exercise 5.13 . . . 11
Exercise 5.14 . . . 11
Exercise 5.15 . . . 11
Exercise 5.16 . . . 12
Exercise 5.17 . . . 15
Exercise 5.18 . . . 15
Exercise 5.19 . . . 15
Exercise 5.20 . . . 16
Exercise 5.21 . . . 16
Exercise 5.22 . . . 16
Exercise 5.23 . . . 17
Exercise 5.24 . . . 18
Exercise 5.25 . . . 19
Exercise 5.26 . . . 19
Exercise 5.27 . . . 19
Exercise 5.28 . . . 20
Exercise 5.29 . . . 20
Exercise 5.30 . . . 21
Exercise 5.31 . . . 21
Exercise 5.32 . . . 22
Exercise 5.33 . . . 22
Exercise 5.34 . . . 22
Exercise 5.35 . . . 23
Exercise 5.36 . . . 23
Exercise 5.37 . . . 23
Exercise 5.38 . . . 24
Exercise 5.39 . . . 24
Exercise 5.40 . . . 24
Exercise 5.41 . . . 25
Exercise 5.42 . . . 25
Exercise 5.43 . . . 26
Exercise 5.44 . . . 26
Exercise 5.45 . . . 27

Exercise 5.46 . . . 27
Exercise 5.47 . . . 27
Exercise 5.48 . . . 28
Exercise 5.49 . . . 29
Exercise 5.50 . . . 30
Exercise 5.51 . . . 30
Exercise 5.52 . . . 30
Exercise 5.53 . . . 32
Exercise 5.54 . . . 32
Exercise 5.55 . . . 32
Exercise 5.56 . . . 33
Exercise 5.57 . . . 33
Exercise 5.58 . . . 34
Exercise 5.59 . . . 35
Exercise 5.60 . . . 35
Exercise 5.61 . . . 36
Exercise 5.62 . . . 37
Exercise 5.63 . . . 38
Exercise 5.64 . . . 39
Exercise 5.65 . . . 39
Exercise 5.66 . . . 40
Exercise 5.67 . . . 41

# Chapter 5 : Elements of Functional Analysis

**Exercise 5.1.** If $X$ is a normed vector space over $\mathbf{K}(=\mathbf{R}$ or $\mathbf{C})$, then addition and scalar multiplication are continuous from $X \times X$ and $\mathbb{K} \times X$ to $X$. Moreover, the norm is continuous from $X$ to $[0, \infty)$; in fact, $|\|x\| - \|y\|| \leq \|x - y\|$.

*Solution.* We will consider the product norm on the product spaces. To prove continuity of addition notice that if, $\|(a, b) - (c, d)\|_{X\times Y} = \max\{\|a - c\|_X, \|b - d\|_X\} < \epsilon$ then

$$\|(a + b) - (c + d)\|_X \leq \|a - c\|_X + \|b - d\|_X \leq 2 \cdot \|(a, b) - (c, d)\|_{X\times X} < 2\epsilon$$

Similary to see the continuity of scalar multiplication, say $\|(\alpha, x) - (\beta, y)\|_{\mathbb{K}\times X} < \epsilon$ then,

$$\|\alpha x - \beta y\|_X = \|\alpha(x - y) + (\alpha - \beta)y\|_X \leq |\alpha|\|x - y\|_X + |\alpha - \beta|\|y\|_X < \epsilon(|\alpha| + \|y\|_X)$$

which proves continuity. The reverse triangle inequality directly proves the continuity of the norm so we prove only that

$$\|x\| \leq \|y\| + \|x - y\| \implies \|x\| - \|y\| \leq \|x - y\|$$

and the same works when we switch $x, y$ proving the reverse triangle inequality. □

**Exercise 5.2.** $L(X, Y)$ is a vector space and the function $\|\cdot\|$ defined by (5.3) is a norm on it. In particular, the three expressions on the right of (5.3) are always equal.

*Solution.* The vector space axioms are easily verified and for $S, T \in L(X, Y)$ and $x \in X$ we have

$$\|(S + T)x\|_Y = \|Sx + Tx\|_Y \leq \|Tx\|_Y + \|Sx\|_Y \leq (\|T\| + \|S\|)\|x\|_X$$

and $\|T\| < \infty$ by definition. If we have $\|T\| = 0$ then $\|Tx\|_Y \neq 0 \cdot \|x\|_X$ and thus $Tx = 0$ for all $x$ implying that $T = 0_{L(X,Y)}$. The first two among the three experssions in (5.3) are equal because the set we take the supremum over is the same, indeed if $x \neq 0$ then $\|x/\|x\|\| = 1$ and

$$\frac{\|Tx\|}{\|x\|} = \frac{\|T(x/\|x\|)\|}{\|x/\|x\|\|}$$

By the definition of the infimum we see that the second is atleast as large as the third and if the third is $c$ then we see that for all $\epsilon > 0$ we have $\|Tx\| \leq (c + \epsilon)\|x\|$ implying $c + \epsilon$ is atleast as large as the second quantity, proving their equality due to the arbitrage of $\epsilon$. □

**Exercise 5.3.** Complete the proof of Proposition 5.4.

*Solution.* As $\{T_n\}$ is cauchy, for any $\epsilon > 0$ we can find $N$ such that $m, n \geq N$ implies $\|T_m - T_n\| < \epsilon$ and thus for all $x \in X$ we have $\|T_m x - T_n x\| \leq \epsilon\|x\|$ and

$$\lim_{m\to\infty} \frac{\|T_m x - T_n x\|}{\|x\|} = \frac{\|Tx - T_n x\|}{\|x\|} \leq \epsilon \quad (\text{By continuity of norms as } \lim T_n x = Tx)$$

implying $\|T_n - T\| \to 0$ as $\epsilon$ was arbitrary, from which $\lim \|T_n\| = \|T\|$ follows from triangle inequality. □

**Exercise 5.4.** If $X, Y$ are normed vector spaces, the map $(T, x) \mapsto Tx$ is continuous from $L(X, Y) \times X$ to $Y$. (That is, if $T_n \to T$ and $x_n \to x$ then $T_n x_n \to Tx$.)

*Solution.* For any $\epsilon > 0$ we can find $N$ such that $n \geq N$ implies $\|T_n - T\|, \|x_n - x\| < \epsilon$ and thus

$$\|T_n x_n - Tx\| \leq \|T_n\|\|x_n - x\| + \|T_n - T\|\|x\| < \epsilon(\|T_n\| + \|x\|)$$

proving $T_n x_n \to Tx$ as $\|T_n\|$ is bounded, being a convergent sequence (Exercise 5.3.) $\square$

**Exercise 5.5.** If $X$ is a normed vector space, the closure of any subspace of $X$ is a subspace.

If $x_n \to x, y_n \to y$ in $X$ then $x_n + y_n \to x + y$ by continuity of addition and for $\alpha \in \mathbb{K}$, $\alpha x_n \to \alpha x$ by continuity of scalar multiplication so the subspace axioms are sufficed for the closure as $x_n + y_n, \alpha x_n$ are sequences in the original subspace.

**Exercise 5.6.** Suppose that $X$ is a finite-dimensional vector space. Let $e_1, \ldots, e_n$ be a basis for $X$, and define $\| \sum_1^n a_j e_j \|_1 = \sum_1^n |a_j|$.

a. $\| \cdot \|_1$ is a norm on $X$.

b. The map $(a_1, \ldots, a_n) \mapsto \sum_1^n a_j e_j$ is continuous from $\mathbb{K}^n$ with the usual Euclidean topology to $X$ with the topology defined by $\| \cdot \|_1$.

c. $\{x \in X : \|x\|_1 = 1\}$ is compact in the topology defined by $\| \cdot \|_1$.

d. All norms on $X$ are equivalent. (Compare any norm to $\| \cdot \|_1$.)

*Solution.*

a. This is trivially verified.

b. The usual topology is generated by the usual euclidean norm. As this is linear, it suffices to prove boundedness:

$$\left\| \sum_{j=1}^n a_j e_j \right\|_1 = \sum_{j=1}^n |a_j| \leq n^{1/2} \cdot \sqrt{\sum_{j=1}^n |a_j|^2} \quad \text{(by Cauchy Schwarz)}$$

c. As $\{(a_1, \ldots, a_n) \in \mathbb{K}^n : \sum_1^\infty |a_j| = 1\}$ is compact in $\mathbb{K}^n$ (being closed and bounded) and mapping in part (b) is continuous, this too must be compact.

d. Let the compact set in part (b) be $B$. Let $\|\cdot\|$ be any norm on $X$, this is continuous with respect to the 1-norm as

$$\left\| \sum_{k=1}^n a_k e_k \right\| \leq \sum_{k=1}^n |a_k|\|e_k\| \leq \max_{1 \leq j \leq n} \|e_j\| \cdot \left\| \sum_{k=1}^n a_k e_k \right\|_1$$

then as continuous functions attain their extremums on compact sets, we can find $p, q \in B$ such that $\sup_{x \in B} \|x\| = \|p\|$ and $\inf_{x \in B} \|x\| = \|q\|$, both of these are

strictly positive and finite by the definition of a norm. Let this supremum be $C$ and infimum $c$, then as $x/\|x\|_1 \in B$ for all $x \in X \setminus \{0\}$ we see that

$$c\|x\|_1 \leq \|x\| = \|x\|_1 \cdot \left\| \frac{x}{\|x\|_1} \right\| \leq C\|x\|_1$$

proving that all norms are equivalent to $\|\cdot\|_1$ and thus they are all mutually equivalent too as equivalence of norms is clearly an equivalence relation.

□

**Exercise 5.7.** Let $X$ be a Banach space.

a. If $T \in L(X,X)$ and $\|I - T\| < 1$ where $I$ is the identity operator, then $T$ is invertible; in fact, the series $\sum_0^\infty (I-T)^n$ converges in $L(X,X)$ to $T^{-1}$.

b. If $T \in L(X,X)$ is invertible and $\|S - T\| < \|T^{-1}\|^{-1}$, then $S$ is invertible. Thus the set of invertible operators is open in $L(X,X)$.

*Solution.*

a. As $X$ is a banach space so is $L(X,X)$ and thus the series $\sum_0^\infty (I-T)^n$ converges in $L(X,X)$ given absolute convergence (this is evident from the fact that $\|I-T\| < 1$.) Thus, $T\sum_0^m (I-T)^n = (I-(I-T))\sum_0^m (I-T)^n = I - (I-T)^{m+1} \to I$ as $\|I-T\|^m \to 0$. This proves that $\sum_0^\infty (I-T)^n = T^{-1}$ as, if $S_n \to S$ in $L(X,X)$ then clearly $TS_n \to TS$ too (here, take $S_n$ to be the $n$-th partial sum of this series. As $TS_n \to I$ we must have $TS = I$ and $ST = I$ follows in the exact same manner as everything commutes nicely.)

b. As $S - T, T \in L(X,X)$ we clearly have $S \in L(X,X)$ and as

$$\|I - T^{-1}S\| = \|T^{-1}T(I - T^{-1}S)\| \leq \|T^{-1}\|\|T - S\| < 1$$

proving $T^{-1}S$ to be invertible by part (a) which in turn makes $S$ invertible, being a composition of invertible maps $S = T \circ (T^{-1}S)$. Now, or any $x \in X$ we see that $\|Sx - Tx\| \leq \|(S-T)T^{-1}Tx\| \leq \|S-T\|\|T^{-1}\|\|Tx\| = C\|Tx\|$ where $C < 1$ by hypothesis. Thus, $\|Sx\| \geq \|Tx\| - \|Sx - Tx\| \geq (1-C)\|Tx\| \geq (1-C)\|T^{-1}\|^{-1}\|x\|$ proving that $S \in L(X,X)$.

□

**Exercise 5.8.** Let $(X, \mathcal{M})$ be a measurable space, and let $M(X)$ be the space of complex measures on $(X, \mathcal{M})$. Then $\|\mu\| = |\mu|(X)$ is a norm on $M(X)$ that makes $M(X)$ into a Banach space. (Use Theorem 5.1.)

*Solution.* If $\|\mu\| = |\mu|(X) = 0$ then we see that for all $E \in \mathcal{M}$ we have $|\mu(E)| \leq |\mu|(E) \leq |\mu|(X) = 0$ making $\mu$ the 0 measure. It can be easily shown that $|\mu_1 + \mu_2| \leq |\mu_1| + |\mu_2|$, which proves the triangle inequality and for scalars $\alpha \in \mathbb{K}$ we clearly see that $|\alpha \cdot \mu| = |\alpha| \cdot |\mu|$ so this is indeed a norm. To show completeness we use the theorem mentioned in the theorem which shows us that it suffices to prove that $\sum_1^\infty \mu_j$ converges if $\sum_1^\infty \|\mu_j\| < \infty$ (this is clearly a positive measure.) If this was the case then we first show that $\sum_1^\infty \mu_j$ is a complex measure. Clearly $\mu_j \ll \sum_1^\infty |\mu_i| =$ say, $\nu$ and we can find

radon nikodym derivatives $f_j$ such that $d\mu_j = f_j d\nu$ and by hypothesis $\sum_1^\infty |f_j| \in L^1(\nu)$ and by DCT, for all $E \in \mathcal{M}$

$$\int_E \sum_{j=1}^\infty f_j d\nu = \sum_{j=1}^\infty \int_E f_j d\nu = \left(\sum_{j=1}^\infty \mu_j\right)(E)$$

Now, as $\sum_1^\infty f_j \in L^1(\nu), \sum_1^\infty \mu_j$ is a complex measure. We have $\sum_{j>n} \mu_j = \sum_{j>n} f_j d\nu$ and just like above we have $\sum_{j>n} f_j d\nu = \left(\sum_{j>n} f_j\right) d\nu$ by DCT and as $\left|\sum_{j>n} f_j\right| \leq \sum_{j>n} |f_j|$ we see that $\left\|\sum_{j>n} \mu_j\right\| \leq \sum_{j>n} \|\mu_j\| \to 0$. □

**Exercise 5.9.** Let $C^k([0,1])$ be the space of functions on $[0,1]$ possessing continuous derivatives up to order $k$ on $[0,1]$, including one-sided derivatives at the endpoints.

a. If $f \in C([0,1])$, then $f \in C^k([0,1])$ iff $f$ is $k$ times continuously differentiable on $(0,1)$ and $\lim_{x \searrow 0} f^{(j)}(x)$ and $\lim_{x \nearrow 1} f^{(j)}(x)$ exist for $j \leq k$. (The mean value theorem is useful.)

b. $\|f\| = \sum_0^k \|f^{(j)}\|_u$ is a norm on $C^k([0,1])$ that makes $C^k([0,1])$ into a Banach space. (Use induction on $k$. The essential point is that if $\{f_n\} \subset C^1([0,1])$, $f_n \to f$ uniformly, and $f_n' \to g$ uniformly, then $f \in C^1([0,1])$ and $f' = g$. The easy way to prove this is to show that $f(x) - f(0) = \int_0^x g(t)\, dt$.)

*Solution.*

a. ( $\implies$ ) This is trivial as differentiablity implies continuity (the one sided limits work out nicely too.)
( $\impliedby$ ) For $j \leq k$ and $x_n \searrow 0$ in $(0,1)$ we can find $y_n \in (0, x_n)$ such that

$$\lim_{x_n \searrow 0} \frac{f^{(j-1)}(x_n) - f^{(j-1)}(0)}{x - 0} = \lim_{x_n \searrow 0} f^{(j)}(y_n) \quad \text{which exists}$$

by the mid value theorem, this proves the existence of the one sided deirvative at 0 by the sequential criterion, that at 1 follows similarly.

b. This is trivially a vectorspace. Clearly $f \mapsto \|f\|_u$ is a norm and $f \mapsto \|f^{(j)}\|_u$ a seminorm for $j = 1, \ldots, k$ so their pointwise sum is clearly a norm. If $f_n$ is cauchy in this norm then we see that $f_n^{(j)}$ are also cauchy in the uniform norm and as $C([0,1])$ is complete in the uniform norm, $f_n^{(j)} \to g_j$ uniformly where $g_j \in C([0,1])$. We will show that $g_0^{(j)} = g_j$ which proves the claim. As $f_n' \to g_1$ uniformly we see that $g_0(x) - g_0(0) = \lim_n f_n(x) - f_n(0) = \lim_n \int_0^x f_n' = \int_0^x g_1$ and thus by FTC $g_0' = g_1$ and we can proceed similarly by induction to prove that $g_0^{(j)} = g_j$ and thus $f_n \to g_0$ in this norm with $g_0 \in C^k([0,1])$.

□

**Exercise 5.10.** Let $L_k^1([0,1])$ be the space of all $f \in C^{k-1}([0,1])$ such that $f^{(k-1)}$ is absolutely continuous on $[0,1]$ (and hence $f^{(k)}$ exists a.e. and is in $L^1([0,1])$). Then $\|f\| = \sum_0^k \int_0^1 |f^{(j)}(x)|\, dx$ is a norm on $L_k^1([0,1])$ that makes $L_k^1([0,1])$ into a Banach space. (See Exercise 9 and its hint.)

*Solution.* This is trivially a vectorspace. $f \mapsto \int_0^1 |f|$ is a norm as the $f$ are continuous here and $f \mapsto \int_0^1 |f^{(j)}|$ is a seminorm for $j = 1, \ldots, k$ so the sum of these is a norm. We will prove this for the case $k = 1$ first using theorem 5.1. Say $f_n \in L_1^1$ such that $\sum \|f_n\| = \sum \int |f_n| + \sum \int |f_n'| = \int \sum |f_n| + \int \sum |f_n'| < \infty$ (note that if $f_n'$ are defined a.e. each, then they are defined all together a.e. too.) Thus, $\sum |f_n|, \sum |f_n'|$ are finite a.e. Suppose $x \in [0,1]$ and that $\sum |f_n(p)|$ is finite for $p \in [0,1] \setminus \{x\}$ (we can always pick such $p$ by the previous line)

$$\sum |f_n(x)| = \sum \left| f_n(p) + \int_p^x f_n' \right| \leq \sum |f_n(p)| + \sum \int_p^x |f_n'| < \infty$$

proving that $\sum |f_n|$ converges everywhere and thus $\sum f_n$ does too, to say $f$. Then, for all $x \in [0,1]$, as $\sum |f_n'|$ is finite a.e. we see that $\sum f_n'$ converges a.e. to say $h$ (if the absolute value sum converges at some point, then this converges too by completness of $\mathbb{C}$) and is integrable (it is defined where it converges and is thus measurable on this set) by the triangle inequality, so

$$f(x) - f(0) = \sum f_n(x) - f_n(0) = \sum \int_0^x f_n' = \int_0^x h$$

where we use DCT in the third equality, this proves that $f$ is absolutely continuous and $f' = h$ exists a.e. and is integrable. We claim that $\sum_1^n f_j \to f$. We see that,

$$\left\| f - \sum_1^n f_j \right\| = \int_0^1 \left| f - \sum_1^n f_j \right| + \int_0^1 \left| h - \sum_1^n f_j' \right|$$

And as $\int_0^1 \sum |f_n| < \infty$, the first term tends to 0 by DCT (keep in mind that DCT produces a conclusion stronger than what is usually considered, i.e. $L^1$ convergence.) As $\int \sum |f_n'| < \infty$ we also have that $\sum f_n' \to h$ in $L^1$ using DCT so we are done. Now, for the general case of $f_n \in L_k^1$ with $\sum_1^\infty \|f_n\| < \infty$ we see that $f_n^{(k-1)} \in L_1^1$ and $\sum \|f^{(k-1)}\|_{L_1^1} < \infty$ too and thus we can get the absolute continuity of $\sum f^{(k-1)}$ following the above arguemt and the rest follows similarly. □

**Exercise 5.11.** If $0 < \alpha \leq 1$ let $\Lambda_\alpha([0,1])$ be the space of Hölder continuous functions of exponent $\alpha$ on $[0,1]$. That is, $f \in \Lambda_\alpha([0,1])$ iff $\|f\|_{\Lambda_\alpha} < \infty$, where

$$\|f\|_{\Lambda_\alpha} = |f(0)| + \sup_{x,y \in [0,1], x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}.$$

a. $\| \cdot \|_{\Lambda_\alpha}$ is a norm that makes $\Lambda_\alpha([0,1])$ into a Banach space.

b. Let $\lambda_\alpha([0,1])$ be the set of all $f \in \Lambda_\alpha([0,1])$ such that $\frac{|f(x)-f(y)|}{|x-y|^\alpha} \to 0$ as $x \to y$, for all $y \in [0,1]$. If $\alpha < 1$, $\lambda_\alpha([0,1])$ is an infinite-dimensional closed subspace of $\Lambda_\alpha([0,1])$. If $\alpha = 1$, $\lambda_\alpha([0,1])$ contains only constant functions.

*Solution.*

a. This is clearly a seminorm, to prove definiteness, observe that $\|f\|_{\Lambda_\alpha} = 0$ implies $f(0) = 0$ and $|f(x) - f(0)| = |f(x)| \leq 0 \cdot |x - 0|^\alpha = 0$ and thus $f = 0$ everywhere.

Now, suppose $f_n \in \Lambda_\alpha([0,1])$ form a cauchy sequence, then for $\epsilon > 0$ we can find $N$ such that $m, n \geq N$ implies $\|f_n - f_m\|_{\Lambda_\alpha} < \epsilon$ and thus

$$|f_m(x) - f_n(x)| \leq |(f_m - f_n)(x) - (f_m - f_n)(0)| + |(f_m - f_n)(0)| \leq \epsilon |x - 0|^\alpha + \epsilon \leq 2\epsilon$$

as $|x|^\alpha \leq 1$ here and thus $\{f_m(x)\}$ are cauchy sequences too and hence they converge. So we can find $f$ such that $f_m \to f$ pointwise, we will now prove that $f \in \Lambda_\alpha([0,1])$ and the fact that $\|f_n - f\|_{\Lambda_\alpha} \to 0$. Now, $\left\{\|f_m\|_{\Lambda_\alpha}\right\}$ is a bounded sequence, as $\{f_m\}$ was cauchy and we can write

$$|f_m(x) - f_m(y)| \leq \|f_m\| \, |x - y|^\alpha \leq M |x - y|^\alpha$$

where $\|f_m\|_{\Lambda_\alpha} \leq M$ and taking the limits as $m \to \infty$ we see that $|f(x) - f(y)| \leq M|x - y|^\alpha$ too and thus $\|f\|_{\Lambda_\alpha} \leq |f(0)| + M < \infty$ proving $f \in \Lambda_\alpha([0,1])$. For arbitrary $\epsilon > 0$, by cauchy property, for large enough $m, n$ we have

$$|(f_m - f_n)(x) - (f_m - f_n)(y)| \leq \epsilon |x - y|^\alpha$$

and taking the limit as $m \to \infty$ we see that

$$\frac{|(f - f_n)(x) - (f - f_n)(y)|}{|x - y|^\alpha} \leq \epsilon \implies \|f - f_n\|_{\Lambda_\alpha} \leq \epsilon + |f_n(0) - f(0)|$$

for large enough $n$ and as $\epsilon$ was arbitrary and $f_n(0) \to f(0)$, we must have $\|f - f_n\|_{\Lambda_\alpha} \to 0$ too proving that this is a banach space.

b. Say $f_n \in \lambda_\alpha([0,1])$ is a convergent sequence in $\Lambda_\alpha([0,1])$, to say $f$. Fix any $y$, if we have $\|f - f_n\|_{\Lambda_\alpha} < \epsilon$ and $|x - y| < \delta$ implies $|f_n(x) - f_n(y)| \leq \epsilon |x - y|^\alpha$ then,

$$|f(x) - f(y)| \leq |(f - f_n)(x) - (f - f_n)(y)| + |f_n(x) - f_n(y)| \leq \epsilon |x - y|^\alpha + \epsilon |x - y|^\alpha$$

proving that $f \in \lambda_\alpha([0,1])$ too. If $\alpha < 1$ then,

$$\frac{f(x) - f(y|}{|x - y|^\alpha} = \frac{|f(x) - f(y)|}{|x - y|} \cdot |x - y|^{1-\alpha}$$

so clearly if $f$ is differentiable then as $x \to y$ this tends $|f'(y)| \cdot 0 = 0$ so this is clearly infinite dimensional as we can find an infinite linearly independant set like the set of all monomials. If $\alpha = 1$ then the definition implies that the derivative is 0 everywhere so the functions have to be constant.

□

**Exercise 5.12.** Let $X$ be a normed vector space and $\mathcal{M}$ a proper closed subspace of $X$.

a. $\|x + \mathcal{M}\| = \inf\{\|x + y\| : y \in \mathcal{M}\}$ is a norm on $X/\mathcal{M}$.

b. For any $\epsilon > 0$ there exists $x \in X$ such that $\|x\| = 1$ and $\|x + \mathcal{M}\| \geq 1 - \epsilon$.

c. The projection map $\pi(x) = x + \mathcal{M}$ from $X$ to $X/\mathcal{M}$ has norm 1.

d. If $X$ is complete, so is $X/\mathcal{M}$. (Use Theorem 5.1.)

e. The topology defined by the quotient norm is the quotient topology as defined in Exercise 28 in §4.2.

*Solution.*

a. This is clearly well defined as if $x - y = z \in \mathcal{M}$ then

$$\|x + \mathcal{M}\| = \inf\{\|y + z + v\| : v \in \mathcal{M}\} = \inf\{\|y + u\| : u \in \mathcal{M}\} = \|y + \mathcal{M}\|$$

as $\{z + v : v \in \mathcal{M}\} = \mathcal{M}$ due to it being a subspace. Now, if $y \in \mathcal{M}$ iff $-y \in \mathcal{M}$ we find that $\|x + \mathcal{M}\| = d(x, \mathcal{M})$ in previously used notation where $d$ is the metric induced by $\|\cdot\|$ and we have proved in Exercise 4.3 that $d(x, \mathcal{M}) = 0$ iff $x \in \mathcal{M}$ so we have definiteness. If $\alpha \in \mathbb{K}$ and $\alpha = 0$ then clearly $\|\alpha x + \mathcal{M}\| = |\alpha| \|x + \mathcal{M}\| = 0$ and otherwise as $y \in \mathcal{M}$ iff $\alpha y \in \mathcal{M}$ we have $\|\alpha x + \mathcal{M}\| = \inf\{|\alpha| \cdot \|x + \alpha^{-1} y\| : y \in \mathcal{M}\} = |\alpha| \cdot \inf\{\|x + y\| : y \in \mathcal{M}\} = |\alpha| \cdot \|x + \mathcal{M}\|$. Now, say $x_n, y_n \in \mathcal{M}$ are such that $\|x + x_n\| \to \|x + \mathcal{M}\|$ and $\|y + y_n\| \to \|y + \mathcal{M}\|$. Then, as $x_n + y_n \in \mathcal{M}$ too we have that, $\|x + y + \mathcal{M}\| \leq \|x + y + x_n + y_n\| \leq \|x + x_n\| + \|y + y_n\| \to \|x + \mathcal{M}\| + \|y + \mathcal{M}\|$ and thus

$$\|x + y + \mathcal{M}\| \leq \|x + \mathcal{M}\| + \|y + \mathcal{M}\|$$

proving the triangle inequality.

b. Consider $v \notin \mathcal{M}$ and then for all $\epsilon > 0$ we can find $y \in \mathcal{M}$ such that $\|v + \mathcal{M}\| > \|v + y\| - \epsilon$. Thus,

$$\left\| \frac{v + y}{\|v + y\|} + \mathcal{M} \right\| = \frac{1}{\|v + y\|} \cdot \|v + \mathcal{M}\| > 1 - \frac{\epsilon}{\|v + y\|} \geq 1 - \frac{\epsilon}{\|v + \mathcal{M}\|}$$

and as $\epsilon > 0$ was arbitrary we are done.

c. As $0 \in \mathcal{M}$ we clearly have $\|x\| \geq \|x + \mathcal{M}\|$, using part (b) we see that

$$\|\pi\| = \sup_{\|x\|=1} \|x + \mathcal{M}\| = 1$$

d. Suppose we have $\sum_1^\infty \|x_n + \mathcal{M}\| < \infty$, then we can find $y_n \in \mathcal{M}$ such that $\|x_n + \mathcal{M}\| \geq \|x_n + y_n\| - 2^{-n}$ and thus $\sum_1^\infty \|x_n + y_n\| < \infty$ so $\sum_1^\infty (x_n + y_n)$ converges. By part (c), we have that $\pi \in L(X, X/\mathcal{M})$ and thus

$$\sum_{j=1}^n (x_j + \mathcal{M}) = \pi\left(\sum_{j=1}^n (x_j + y_j)\right) \to \pi\left(\sum_{j=1}^\infty (x_j + y_j)\right)$$

proving convergence, and thus by Theorem 5.1 we are done.

e. As $\pi$ is continuous when $X/\mathcal{M}$ has its topology generated by the quotient norm, we see that $\pi^{-1}(U)$ is open in $X$ for all open $U \subseteq X/M$ and if $\pi^{-1}(U) \subseteq X$ is open then we want to show that $U$ is open in $X/\mathcal{M}$ with the quotient norm topology. Say $u + \mathcal{M} \in U$, then we can find $x \in X$ such that $\pi(x) = u + \mathcal{M}$, now we can find some $\delta > 0$ such that $B_X(x, \delta) \subseteq \pi^{-1}(U)$ and if $\|x - y\| < \delta$ then $\|\pi(x - y)\| \leq \|\pi\| \|x - y\| = \|x - y\| < \delta$ so $\|(x + \mathcal{M}) - (y + \mathcal{M})\| < \delta$. If

$\|x - y + \mathcal{M}\| < \delta/2$ then we can find $z \in \mathcal{M}$ with $\|x - y + z\| < \delta$ and thus $(y - z) \in B_X(x, \delta)$ implying $\pi(y - z) = y + \mathcal{M} \in \pi(B_X(x, \delta))$

$$\left\{ y \in \mathcal{M} : \|x - y + \mathcal{M}\| < \frac{\delta}{2} \right\} \subseteq \pi(B_X(x, \delta)) \subseteq \pi\Big(\pi^{-1}(U)\Big) \subseteq U$$

proving that it is open in the quotient norm topology. Infact, instead of $\delta/2$ any positive real strictly less than $\delta$ works too so $\delta$ itself works here. We have thus proved that the quotient norm topology is finer than the quotient topology and we proved the other direction of the inclusion at the start so they must be the same topology.

□

**Exercise 5.13.** If $\|\cdot\|$ is a seminorm on the vector space $X$, let $\mathcal{M} = \{x \in X : \|x\| = 0\}$. Then $\mathcal{M}$ is a subspace, and the map $x + \mathcal{M} \mapsto \|x\|$ is a norm on $X/\mathcal{M}$.

*Solution.* We claim that this map is the same as $x + \mathcal{M} \mapsto \|x + \mathcal{M}\|$ in the notation of Exercise 5.12(part (a) specifically). If $\|z\| = 0$ then $\|x\| = |\,\|x\| - \|z\|\,| \leq \|x + z\| \leq \|x\| + \|z\| = \|x\|$ and thus $\inf_{z \in \mathcal{M}} \|x + z\| = \|x\|$ so this the quotient norm induced by $\mathcal{M}$ and we are done. □

**Exercise 5.14.** If $X$ is a normed vector space and $\mathcal{M}$ is a nonclosed subspace, then $\|x + \mathcal{M}\|$, as defined in Exercise 12, is a seminorm on $X/\mathcal{M}$. If one divides by its nullspace as in Exercise 13, the resulting quotient space is isometrically isomorphic to $X/\overline{\mathcal{M}}$. (Cf. Exercise 5.)

*Solution.* Let $\mathcal{N} := \{x + \mathcal{M} \in X/\mathcal{M} : \|x + \mathcal{M}\| = 0\}$, we want to prove that $(X/\mathcal{M})/\mathcal{N}$ is isometrically isomorphic to $X/\overline{\mathcal{M}}$. By Exercise 5.13 we see that

$$(x + \mathcal{M}) + \mathcal{N} \mapsto \|x + \mathcal{M}\|$$

is a norm on $(X/\mathcal{M})/\mathcal{N}$. Now, $x + \overline{\mathcal{M}} \mapsto \|x + \overline{\mathcal{M}}\|$ is the induced norm on $X/\overline{\mathcal{M}}$ and we claim that the map

$$(x + \mathcal{M}) + \mathcal{N} \mapsto x + \overline{\mathcal{M}}$$

is well defined and an isometric isomorphism. To prove well definedness, we first pick two representatives $x, y$ with $(x + \mathcal{M}) + \mathcal{N} = (y + \mathcal{M}) + \mathcal{N}$, then we must have that $(x + \mathcal{M}) - (y + \mathcal{M}) = x - y + \mathcal{M} = z + \mathcal{M}$ for some $z + \mathcal{M} \in \mathcal{N}$ and thus, $\|x - y + \mathcal{M}\| = 0 = \inf_{v \in \mathcal{M}} \|x - y - v\|$ which implies that $x - y \in \overline{\mathcal{M}}$ and hence $x + \overline{\mathcal{M}} = y + \overline{\mathcal{M}}$ proving well definedness. This map is clearly linear, surjective by construction. We now prove that $\|x + \mathcal{M}\| = \|x + \overline{\mathcal{M}}\|$. As $\mathcal{M} \subseteq \overline{\mathcal{M}}$, the left side is clearly atleast as large and the right. Now, for any $\epsilon > 0$ we can find $y \in \overline{\mathcal{M}}$ such that $\|x + y\| - \epsilon < \|x + \overline{\mathcal{M}}\|$ and we can also find $v \in mM$ such that $\|y - v\| < \epsilon$ and by the triangle inequality, $\|x + v\| - 2\epsilon < \|x + \overline{\mathcal{M}}\|$ implying that $\|x + \mathcal{M}\| - 2\epsilon \leq \|x + \overline{\mathcal{M}}\|$ and as $\epsilon$ was arbitrary we have $\|x + \mathcal{M}\| \leq \|x + \overline{\mathcal{M}}\|$ so its an isometry, which also proves injectivity and continuity making it an isomorphism too. □

**Exercise 5.15.** Suppose that $X$ and $Y$ are normed vector spaces and $T \in L(X, Y)$. Let $\mathcal{N}(T) = \{x \in X : Tx = 0\}$.

a. $\mathcal{N}(T)$ is a closed subspace of $X$.

b. There is a unique $S \in L(X/\mathcal{N}(T), Y)$ such that $T = S \circ \pi$ where $\pi : X \to X/\mathcal{N}(T)$ is the projection (see Exercise 12). Moreover, $\|S\| = \|T\|$.

*Solution.*

a. $\mathcal{N}(T) = T^{-1}(\{0\})$ is closed as $\{0\}$ is closed (metric spaces are clearly $T_1$) and $T$ continuous.

b. As $\pi$ is a surjection such a map must be unique and if it exists it is given by $x + \mathcal{N}(T) \mapsto T(x)$. This is well defined as $x - y \in \mathcal{N}(T) \implies T(x) = T(y)$ so $S(x + \mathcal{M}) = S(y + \mathcal{M})$. It is clearly linear and by Exercise 4.28 it is continuous as $T = S \circ \pi$ is continuous so $S \in L(X/\mathcal{N}(T), Y)$. Now, if $T$ is not 0 we see that $\mathcal{N}(T)$ is a proper closed subspace and thus, $\|T\| = \|S \circ \pi\| \leq \|S\|$ as $\|\pi\| = 1$ by Exercise 5.12c. Now, for any $\epsilon > 0$ we can find $x + \mathcal{N}(T)$ of unit norm such that $\|S(x + \mathcal{N}(T))\| = \|Tx\| + \epsilon > \|S\|$ and we can also find $v \in \mathcal{N}(T)$ such that $\|x + v\| < 1 + \epsilon$. Thus, as $x + v \neq 0$ clearly (if we had $\|S\| = 0$ then the claim is true already, otherwise for small enough $\epsilon$ we have $\|Tx\| > 0$ so $x \notin \mathcal{N}(T)$ and thus $x + v \neq 0$)

$$\|T\| \geq \frac{\|T(x+v)\|}{\|x+v\|} > \frac{\|S\| - \epsilon}{1+\epsilon}$$

implying $\|T\| > \|S\| - \epsilon(1 + \|T\|)$ and as $\epsilon > 0$ was arbitrary we must have $\|T\| \geq \|S\|$ and we are done showing $\|T\| = \|S\|$ (If $T = 0$ then its trivially true.)

□

**Exercise 5.16.** The purpose of this exercise is to develop a theory of integration for functions with values in a separable Banach space. Let $(X, \mathcal{M}, \mu)$ be a measure space, $Y$ a separable Banach space, and $L_Y$ the space of all $(\mathcal{M}, \mathcal{B}_Y)$-measurable maps from $X$ to $Y$, and $F_Y$ the set of maps $f : X \to Y$ of the form $f(x) = \sum_1^n \chi_{E_j}(x) y_j$ where $n \in \mathbf{N}$, $y_j \in Y$, $E_j \in \mathcal{M}$, and $\mu(E_j) < \infty$. If $f \in L_Y$, since $y \mapsto \|y\|$ is continuous (Exercise 1), $x \mapsto \|f(x)\|$ is $(\mathcal{M}, \mathcal{B}_{\mathbf{R}})$-measurable, and we define $\|f\|_1 = \int \|f(x)\| \, d\mu(x)$. Finally, let $L_Y^1 = \{f \in L_Y : \|f\|_1 < \infty\}$.

a. $L_Y$ is a vector space, $F_Y$ and $L_Y^1$ are subspaces of it, $F_Y \subset L_Y^1$, and $\| \cdot \|_1$ is a seminorm on $L_Y^1$ that becomes a norm if we identify two functions that are equal a.e.

b. Let $\{y_n\}_1^\infty$ be a countable dense set in $Y$. Given $\epsilon > 0$, let $B_n^\epsilon = \{y \in Y : \|y - y_n\| < \epsilon \|y_n\|\}$. Then $\bigcup_1^\infty B_n^\epsilon \supset Y \backslash \{0\}$.

c. If $f \in L_Y^1$ there is a sequence $\{h_n\} \subset F_Y$ with $h_n \to f$ a.e. and $\|h_n - f\|_1 \to 0$. (With notation as in (b), let $A_{nj} = B_n^{1/j} \backslash \bigcup_{m=1}^{n-1} B_m^{1/j}$ and $E_{nj} = f^{-1}(A_{nj})$, and consider $g_j = \sum_{n=1}^\infty y_n \chi_{E_{nj}}$.)

d. There is a unique linear map $\int : L_Y^1 \to Y$ such that $\int y\chi_E = \mu(E)y$ for $y \in Y$ and $E \in \mathcal{M}$ $(\mu(E) < \infty)$, and $\| \int f\| \leq \|f\|_1$.

e. The dominated convergence theorem: If $\{f_n\}$ is a sequence in $L_Y^1$ such that $f_n \to f$, and there exists $g \in L^1$ such that $\|f_n(x)\| \leq g(x)$ for all $n$ and a.e. $x$, then $\int f_n \to \int f$.

f. If $Z$ is a separable Banach space, $T \in L(Y,Z)$, and $f \in L^1_Y$ then $T \circ f \in L^1_Z$ and $\int T \circ f = T(\int f)$.

*Solution.*

a. For $f, g \in L_Y$ let $\Phi : X \to Y \times Y$ be given by $x \mapsto (f(x), g(x))$ and let $\Psi : Y \times Y \to Y$ be the continuous map $(x,y) \mapsto x+y$. As $Y$ is seperable we have that $\mathcal{B}_Y \otimes \mathcal{B}_Y = \mathcal{B}_{Y\times Y}$ (here $Y \times Y$ is given the product norm) by proposition 1.5, thus $\Psi$ is $(\mathcal{B}_Y \otimes \mathcal{B}_Y, \mathcal{B}_Y)$-measurable and as the components of $\Phi$ are measurable, it is $(\mathcal{B}_X, \mathcal{B}_Y \otimes \mathcal{B}_Y)$-measurable and hence so is $\Psi \circ \Phi$ which is given by $x \mapsto f(x)+g(x)$ proving closure under addition. For $\alpha \in \mathbb{K}$, as $x \mapsto \alpha x$ is continuous, composing any $f \in L_Y$ presereves measurablity and thus $\alpha f \in L_Y$ too and thus it is a vectorspace (it is a subspace of the vectorspace $Y^X$ with pointwise addition and scalar multiplication, over $\mathbb{K}$ so we don't have to check all the axioms, this is sufficient.) If $f, g \in L^1_Y$ and $\alpha \in \mathbb{K}$ then,

$$\|f+g\|_1 = \int \|f(x)+g(x)\| d\mu(x) \leq \int \|f(x)\| d\mu(x) + \int \|g(x)\| d\mu(x) = \|f\|_1 + \|g\|_1$$

by properties of the integral for real valued functions and thus $\|f+g\|_1 < \infty$ too. We also have

$$\|\alpha f\| = \int \|\alpha f(x)\| d\mu(x) = |\alpha| \cdot \int \|f(x)\| d\mu(x) = |\alpha| \|f\|_1$$

so $\|\alpha f\|_1 < \infty$ too and we thus have that $L^1_Y$ is a subspace and $\|\cdot\|_1$ is a seminorm on it. For any $\sum_1^n y_j \chi_{E_j} \in F_Y$, define $K = \cup E_j^c \in \mathcal{M}$, then this is constant and hence measurable when restricted to $E_j, K$ and as $K \cup (\cup E_j) = X$ it must be measurable on $X$ by simple induction on Exercise 2.5. $F_Y$ is trivially closed under addition and scalar multiplication making it a subspace of $L_Y$ and the fact that $F_Y \subseteq L^1_Y$ follows directly from the triangle inequality. If we idetify functions a.e. then clearly $\|f\|_1 = 0$ implies $\|f\| = 0$ a.e. i.e. $f = 0$ a.e. so $\|\cdot\|_1$ is a norm in this case.

b. It suffices to prove this for small enough $\epsilon > 0$ so assume $\epsilon < 1$. If $y \neq 0$ then for all *small enough* $\delta$ we can find $n$ with $\|y_n - y\| < \delta$, implying $0 < \|y\| - \delta < \|y_n\|$ and it suffices to have $\delta < \epsilon(\|y\| - \delta)$ rearranging which we get $\delta < (1+\epsilon)^{-1}\epsilon\|y\|$ so $\delta = 2^{-1} \cdot \min\{\|y\|, \epsilon\|y\|/(1-\epsilon)\} > 0$ works and we have $Y \setminus \{0\} \subseteq \cup B_n^\epsilon$.

c. We can assume that all the $y_n$ are nonzero as otherwise we can just remove it and add on countably many points of form $n^{-1}x$ for some $x \neq 0$. Firstly, the $g_j$ in the hint are all measurable using the same arguement we used for functions in $F_Y$ but working with countably many things (it still works out as sigma algebra are closed under countable unions.) Now, by construction we see $\|f(x) - g_j(x)\| = \|f(x) - y_n\| < j^{-1}\|y_n\|$ on $E_{nj}$ and 0 if it isn't in any $E_{nj}$ and we denote this regioin by $S_j^c$ ( $g_j, f$ are both zero here.) Then, for this $x$ we have $\|g_j(x)\| \leq j^{-1}\|g_j(x)\| + \|f(x)\|$ and thus, $\|g_j(x)\| \leq (1-1/j)^{-1}\|f(x)\|$ so we have $\|g_j\| \leq 2\|f\|$ for all $j \geq 2$ and $g_j \in L^1_Y$ for $j \geq 2$. Again, for this $x$ we have $\|f(x) - g_j(x)\| < j^{-1}(1-j^{-1})^{-1}\|f(x)\| = (j-1)^{-1}\|f(x)\|$ so we find that $g_j \to f$ pointwise and as both of these are $L^1_Y$, by DCT we have $\|f - g_j\|_1 \to 0$ ( $\|f - g_j\| \leq 3\|f\|$ for $j \geq 2$) but $g_j$ need not be in $F_Y$. Now, let $h_{mj} := \sum_{n=1}^n y_n \chi_{E_{nj}}$, these are in $F_Y$

for $n \geq 2$ (as $g_j$is $L^1_Y$ for $n \geq 2$ we have $\mu(E_{kj}) < \infty$ for $j \geq 2$ as $y_k$ are all nonzero.) Clearly $\lim_m h_{mj} = g_j$ and we also have convergence in $L^1_Y$ by DCT. Thus, we can find $m_j$ such that $\|h_{m_j j} - g_j\|_1 < (j-1)^{-1}$ and by previous work we have $\|h_{m_j j} - f\|_1 \leq \|h_{m_j j} - g_j\|_1 + \|g_j - f\|_1 \leq (j-1)^{-1}(1 + \|f\|_1) \to 0$ so the sequence $h_n := h_{m_n n} \in F_Y$ converges to $f$ in $L^1_Y$. Now, as $\|h_n - f\| \to 0$ in $L^1$, by Corollary 2.32 we can find $h_{n_k} \to f$ a.e. and clearly $h_{n_k}$ converges to $f$ in $L^1_Y$ too so we are done.

d. For any $f \in L^1_Y$, by the previous work there exists $h_n \in F_Y$ such that $h_n \to f$ in $L^1_Y$ and a.e. too. First, let $\int$ be defined on $F_Y$ by $\int \sum_1^m y_j \chi_{E_j} := \sum_1^m \mu(E_j) y_j$, this is clearly linear and $\|\int \phi\| \leq \int \|\phi\|$ for $\phi \in F_Y$ follows from the triangle inequality. Now, if $h_n \to f$ as above then consider the sequence $I_n := \int h_n$, we see that $\|I_n - I_m\| = \|\int (h_n - h_m)\| \leq \|h_n - h_m\|_1$ and as $h_n$ is cauchy in $\|\cdot\|_1$ (due to being convergent) we see that $I_n$ is too and thus it converges. We define $\int f := \lim I_n$, this is well defined because if $h_n, k_n \to f$ in that manner then by the triangle inequality $\|h_n - k_n\|_1 \to 0$ and as $\|\int (h_n - k_n)\| \leq \|h_n - k_n\|_1$ we must have that $\lim \int (h_n - k_n) = 0$ so $\lim \int k_n = \lim \int k_n + \lim \int (h_n - k_n) = \lim(\int k_n + \int (h_n - k_n)) = \lim \int h_n$. Now, say $f, g \in L^1_Y$ and $f_n \to f, g_n \to g$ through $F_Y$, in $\|\cdot\|_1$ then we clearly see that $f_n + g_n \to f + g$ through $F_Y$, in $\|\cdot\|_1$ and thus,

$$\int f + g = \lim \int f_n + g_n = \lim \left( \int f_n + \int g_n \right) = \lim \int f_n + \lim \int g_n = \int f + \int g$$

and it can be trivially shown that $\int \alpha f = \alpha \int f$ for $\alpha \in \mathbb{K}$ so this linear. We also see that if $h_n \to f$ similarly then by continuity of the norm(s),

$$\left\| \int f \right\| = \left\| \lim_{n\to\infty} \int h_n \right\| = \lim_{n\to\infty} \left\| \int h_n \right\| \leq \lim_{n\to\infty} \|h_n\|_1 = \left\| \lim_{n\to\infty} h_n \right\|_1 = \|f\|_1$$

Now if $\int^*$ is another linear map as such, then these must agree on $F_Y$ and if $h_n \to f \in L^1_Y$ as above, then

$$\left\| \int^* f - \int h_n \right\| = \left\| \int^* f - \int^* h_n \right\| = \left\| \int^* (f - h_n) \right\| \leq \|f - h_n\|_1 \to 0$$

and thus $\int f = \lim \int h_n = \int^* f$ implying $\int = \int^*$ proving uniqueness.

e. Taking the limits we see that $\|f(x)\| \leq g(x)$ a.e. too and thus $f \in L^1_Y$ as well. Now, $\|\int f - \int f_n\| \leq \|f - f_n\|_1 \to 0$ by the usual DCT as $f - f_n \to 0$ a.e. and $\|f - f_n\| \leq 2g$ a.e. so we have $\int f = \lim \int f_n$.

f. If $T \in L(Y, Z)$ then for $E \in \mathcal{M}$ with $\mu(E) < \infty$ and $y \in Y$, let $f = \chi_E y$ and then $T(f(x)) = \chi_E(x) T(y)$ so $T \circ f \in F_Z$ and we have $\int T \circ f = \int T(y) \chi_E = \chi(E) T(y) = T(\chi(E) y) = T(\int f)$ and we can extend by linearity to $F_Y$. Now, say $f \in L^1_Y$ and $\{h_n\} \subseteq F_Y$ such that $h_n \to f$ a.e. as in part (c), then we see that $T \circ h_n \to T \circ f$ a.e. too and $\|T(h_n(x))\| \leq \|T\| \|h_n(x)\|$. As $T$ is continuous, composition by it preserves measurablity and $\|T(f(x))\| \leq \|T\| \|f(x)|$ so $T \circ f \in L^1_Z$. Now using the generalized DCT in Exercise 2.20 we see that $\int T \circ h_n \to \int T \circ f$ and as $T$ is continuous we have $\int T \circ f = \lim \int T \circ h_n = \lim T(\int h_n) = T(\lim \int h_n) = T(\int f)$ so we are done.

□

**Exercise 5.17.** A linear functional $f$ on a normed vector space $X$ is bounded iff $f^{-1}(\{0\})$ is closed. (Use Exercise 12b.)

If $f$ is bounded then its continuous $f^{-1}(\{0\})$ is closed being the preimage of a closed set. Conversely, using Exercise 5.12b (here the kernel is a closed subspace which we may assume to be proper as otherwise its trivial) we can find $x \in X \setminus \ker(T)$ of norm 1 such that $\|x + \ker(T)\| \geq 1/2$. Consider $Tx \neq 0$, let this be $C$, then we see for all $y$ we have $Ty = T(T(y)C^{-1}x)$ and as $\|T(y)C^{-1}x + \ker(T)\| = \|y + \ker(T)\|$, we have

$$\|y\| \geq \|y + \ker(T)\| \geq (1/2)|T(y)C^{-1}| \implies \frac{|Ty|}{\|y\|} \leq 2C \quad (\text{if } y \neq 0)$$

and we are done.

**Exercise 5.18.** Let $X$ be a normed vector space.

a. If $\mathcal{M}$ is a closed subspace and $x \in X \setminus \mathcal{M}$ then $\mathcal{M} + \mathbf{C}x$ is closed. (Use Theorem 5.8a.)

b. Every finite-dimensional subspace of $X$ is closed.

*Solution.*

a. Let $f$ be as in the proof of Theorem 5.8a. If $x_n + \alpha_n x$ is convergent in $\mathcal{M} + \mathbb{C}x$ with $x_n \in \mathcal{M}$ then $f(x_n + \alpha_n x) = \alpha_n$ also converges by continuity so by alebra of limits we see that $x_n$ converges as well. Now, as $\mathcal{M}$ is closed, $\lim x_n \in X$ and thus $\lim(x_n + \alpha_n x) = \lim x_n + (\lim \alpha_n)x \in \mathcal{M} + \mathbb{C}x$ by continuity of multiplication by scalars as seen in Exercise 5.1 proving that this is closed.

b. We can find linearly independant $x_1, \ldots, x_n$ spanning this subspace. Using the previous part and induction we see that for linearly independant $x_1, \ldots x_n$, the subspaces $\{0\} + \mathbb{C}x_1 = \mathbb{C}x_1, \mathbb{C}x_1 + \mathbb{C}x_2, \ldots, \mathbb{C}x_1 + \ldots + \mathbb{C}x_n$ are all closed so we are done (by linear indepdance we have that $x_k \notin \sum_1^{k-1} \mathbb{C}x_j$ and we could use the previous part.).

□

**Exercise 5.19.** Let $X$ be an infinite-dimensional normed vector space.

a. There is a sequence $\{x_j\}$ in $X$ such that $\|x_j\| = 1$ for all $j$ and $\|x_j - x_k\| \geq \frac{1}{2}$ for $j \neq k$. (Construct $x_j$ inductively, using Exercises 12b and 18.)

b. $X$ is not locally compact.

*Solution.*

a. Following the hint, by the previous problem and Exercise 5.12b we can inductively choose $x_{n+1} \in X \setminus \sum_1^n \mathbb{C}x_j$ such that $\|x_{n+1} - x_k\| \geq \|x_{n+1} - \sum_1^n \mathbb{C}x_j\| \geq 1 - 1/2$ for all $k = 1, \ldots, n$.

b. If it was locally compact then we see that we can find some $\delta > 0$ and compact $K$ such that $B(0, \delta) \subseteq K$ but then this would imply that the sequence $\{\delta x_j\} \subseteq K$ has a converging subsequence (compactness is equivalent to relative compactness) which is absurd as $\|\delta x_i - \delta x_j\| \geq \delta/2$ for $i \neq j$ and hence we have a contradiction.

□

**Exercise 5.20.** If $\mathcal{M}$ is a finite-dimensional subspace of a normed vector space $X$, there is a closed subspace $\mathcal{N}$ such that $\mathcal{M} \cap \mathcal{N} = \{0\}$ and $\mathcal{M} + \mathcal{N} = X$.

*Solution.* Choose a basis $e_1, \ldots, e_k$ of $\mathcal{M}$ and consider linear functionals $f_j$ such that $f_k(e_j) = 1$ and $f_j|_{\text{Span}(\{e_1,\ldots,e_k\}\setminus\{e_j)} = 0$ from theorem 5.8a as all finite dimensional subspaces are closed. Now, we claim that $\mathcal{N} = \cap_1^k \mathcal{N}(f_k)$ works. This is clearly closed and for all $x \in X$,

$$x = \underbrace{\sum_{j=1}^{k} f_j(x)e_j}_{\in\mathcal{M}} + \underbrace{x - \sum_{j=1}^{k} f_j(x)e_j}_{\in\mathcal{N}}$$

and if $z \in \mathcal{M} \cap \mathcal{N}$ then $z = \sum_1^k a_j e_j \implies f_j(z) = a_j = 0$ for all $j$ and thus $z = 0$ so $\mathcal{N}$ works. □

**Exercise 5.21.** If $X$ and $Y$ are normed vector spaces, define $\alpha : X^* \times Y^* \to (X \times Y)^*$ by $\alpha(f, g)(x, y) = f(x) + g(y)$. Then $\alpha$ is an isomorphism which is isometric if we use the norm $\|(x, y)\| = \max(\|x\|, \|y\|)$ on $X \times Y$, the corresponding operator norm on $(X \times Y)^*$, and the norm $\|(f, g)\| = \|f\| + \|g\|$ on $X^* \times Y^*$.

*Solution.* If $\phi \in (X \times Y)^*$ then we see that $\phi(x, y) = \phi(x, 0) + \phi(0, y)$ and clearly $x \mapsto \phi(x, 0)$ is a linear functional on $X$ (proof of continuity is easy) and similarly in the other coordinate. Let these be $\phi_X, \phi_Y$ respectively, then we see that $\alpha(\phi_X, \phi_Y) = \phi$ making it surjective. If $\alpha(f, g)(x, y) = 0$ for all $x, y$ then $\alpha(f, g)(x, 0) = f(x) = 0$ for all $x$ and thus $f = 0$, similarly $g$ is also zero (punctuation!) making it injective. We clearly see that if $\|(x, y)\| = 1$ then $\|\alpha(f, g)(x, y)\| \leq \|f(x)\| + \|g(y)\| \leq \|f\| + \|g\|$ so $\|\alpha(f, g)\| \leq \|(f, g)\|$ proving it to be an isomorphism. Say $\epsilon > 0$, we can find unit $x, y$ with $|f(x)| \geq (1 - \epsilon)\|f\|$ and similarly for $g$ and $y$. If $\alpha := \overline{\operatorname{sgn} f(x)}, \beta = \overline{\operatorname{sgn} g(y)}$ then

$$|\alpha(f, g)(\alpha x, \beta y)| = |f(\alpha x) + g(\beta y)| = ||f(x)| + |g(x)|| \geq (1 - \epsilon)(\|f\| + \|g\|)$$

with $\|(\alpha x, \beta y)\| = 1$ as $\|\alpha x\| = |\alpha|\|x\| = \|x\| = 1$ and similarly for $y$. As $\epsilon > 0$ was arbitrary we can say that $\|\alpha(f, g)\| \geq \|(f, g)\|$ proving that it is an isometry too. □

**Exercise 5.22.** Suppose that $X$ and $Y$ are normed vector spaces and $T \in L(X, Y)$.

a. Define $T^\dagger : Y^* \to X^*$ by $T^\dagger f = f \circ T$. Then $T^\dagger \in L(Y^*, X^*)$ and $\|T^\dagger\| = \|T\|$. $T^\dagger$ is called the adjoint or transpose of $T$.

b. Applying the construction in (a) twice, one obtains $T^{\dagger\dagger} \in L(X^{**}, Y^{**})$. If $X$ and $Y$ are identified with their natural images $\widehat{X}$ and $\widehat{Y}$ in $X^{**}$ and $Y^{**}$, then $T^{\dagger\dagger}|_X = T$.

c. $T^\dagger$ is injective iff the range of $T$ is dense in $Y$.

d. If the range of $T^\dagger$ is dense in $X^*$, then $T$ is injective; the converse is true if $X$ is reflexive.

*Solution.*

a. For all $x$ we have $|T^\dagger(f)(x)| = |f(T(x))| \leq \|T\|\|f\|\|x\|$ so $\|T^\dagger(f)\| \leq \|T\|\|f\|$ and thus $\|T^\dagger\| \leq \|T\|$ proving $T^\dagger \in L(Y^*, X^*)$. For small enough $\epsilon > 0$ we can find unit

norm $x$ such that $\|Tx\| \geq (1-\epsilon)\|T\|\|x\|$ (Assume $\|Tx\| \neq 0$, otherwise its trivial as $T = 0$) and by Theorem 5.8b we can find unit norm $f$ such that $f(Tx) = \|Tx\|$ so $\|T^\dagger(f)(x)\| = \|Tx\| \geq (1-\epsilon)\|T\|$ proving that $\|T^\dagger(f)\| \geq (1-\epsilon)\|T\|$ (remember how $x$ was unit norm) and thus $\|T^\dagger\| \geq (1-\epsilon)\|T\|$ proving the claim via arbitrage of $\epsilon$.

b. This asks us to show that for $x \in X$ we have $T^{\dagger\dagger}(\widehat{x}) = \widehat{Tx}$. Say $f \in Y^*$ then we have $T^{\dagger\dagger}(\widehat{x})(f) = \widehat{x}(T^\dagger(f)) = f(Tx) = \widehat{Tx}(f)$ so we are done.

c. Theorem 5.8a holds even when $\mathcal{M}$ is not closed given that we only have $\|f\| \leq 1$. For any $y \in Y \setminus T(X)$, let $\delta = \inf_{x\in X} \|y - Tx\|$ and we can find $f \in Y^*$ such that $f = 0$ on $T(X)$ and $f(y) = \delta$. If $\delta > 0$ (otherwise $y \in \overline{T(X)}$) then $f \neq 0$ and from injectivity $T^\dagger(f) \neq 0$ so we can find $x$ with $f(Tx) \neq 0$ which is a contradiction so we must have $\delta = 0$ proving $Y \setminus T(X) \subseteq \overline{T(X)}$ and we are done. Conversely, if $T(X)$ is dense in $Y$ and $f \in Y^*$ is nonzero, say $f(y) \neq 0$ then we show that $T^\dagger(f) \neq 0$. For $x \in X$ we have $T^\dagger(f)(x) = f(Tx)$ and if $Tx$ is close enough to $y$ then $f(Tx) \neq 0$ by continuity of $f$ and we can get arbitrarily close by the density assumption making $T^\dagger(f) \neq 0$ proving injectivity.

d. By the previous part, if the range of $T^\dagger$ is dense in $X^*$ then $T^{\dagger\dagger}$ is injective and if we restrict it to $\widehat{X} \subseteq X^{**}$ it retains injectivity. As $X, \widehat{X}$ are isometric under the natural map $x \neq 0$ implies $\|x\| = \|\widehat{x}\| > 0$ and thus $\|T^{\dagger\dagger}(\widehat{x})\| = \|\widehat{Tx}\| = \|Tx\| > 0$ so $Tx \neq 0$ proving injectivity. If $X$ is reflexive then we can identify $X$ with $X^{**}$ via the natural isometry and thus $T$ being injective makes $T^{\dagger\dagger}$ injective, which by the previous part makes $T^\dagger$'s range dense in $X^*$ proving the converse in this case.

□

**Exercise 5.23.** Suppose that $X$ is a Banach space. If $\mathcal{M}$ is a closed subspace of $X$ and $\mathcal{N}$ is a closed subspace of $X^*$, let $\mathcal{M}^0 = \{f \in X^* : f|_{\mathcal{M}} = 0\}$ and $\mathcal{N}^\perp = \{x \in X : f(x) = 0 \text{ for all } f \in \mathcal{N}\}$. (Thus, if we identify $X$ with its image in $X^{**}$, $\mathcal{N}^\perp = \mathcal{N}^0 \cap X$.)

a. $\mathcal{M}^0$ and $\mathcal{N}^\perp$ are closed subspaces of $X^*$ and $X$, respectively.

b. $(\mathcal{M}^0)^\perp = \mathcal{M}$ and $(\mathcal{N}^\perp)^0 \supset \mathcal{N}$. If $X$ is reflexive, $(\mathcal{N}^\perp)^0 = \mathcal{N}$.

c. Let $\pi : X \to X/\mathcal{M}$ be the natural projection, and define $\alpha : (X/\mathcal{M})^* \to X^*$ by $\alpha(f) = f \circ \pi$. Then $\alpha$ is an isometric isomorphism from $(X/\mathcal{M})^*$ onto $\mathcal{M}^0$, where $X/\mathcal{M}$ has the quotient norm.

d. Define $\beta : X^* \to \mathcal{M}^*$ by $\beta(f) = f|_{\mathcal{M}}$; then $\beta$ induces a map $\overline{\beta} : X^*/\mathcal{M}^0 \to \mathcal{M}^*$ as in Exercise 15, and $\overline{\beta}$ is an isometric isomorphism.

*Solution.*

a. These are trivially subspacess. If $f_n \in \mathcal{M}^0$ converge to $f$ then by Exercise 5.4, for any $x \in \mathcal{M}$ we have $f(x) = \lim_n f_n(x) = 0$ proving this to be closed. Now, as $f^{-1}(\{0\})$ is always closed for $f \in X^*$ we see that

$$\mathcal{N}^\perp = \bigcap_{f\in\mathcal{N}} f^{-1}(\{0\})$$

is also closed being an arbitrary intersection of closed sets.

b. If $x \in \mathcal{M}$ and $f \in \mathcal{M}^0$ then $f(x) = 0$ so $x \in (\mathcal{M}^0)^\perp$ so $\mathcal{M} \subseteq (\mathcal{M}^0)^\perp$. If $x \in (\mathcal{M}^0)^\perp$ then $f \in X^*$ being zero on $\mathcal{M}$ makes it zero at $x$. Now, by Theorem 5.8a we can find, for $x \in X \setminus \mathcal{M}$, linear functionals $f \in \mathcal{M}^0$ such that $f(x) \neq 0$ so $\mathcal{M}^c \subseteq ((\mathcal{M}^0)^\perp)^c \implies \mathcal{M} \supseteq (\mathcal{M}^0)^\perp$ so we have set equality. If $f \in \mathcal{N}$ then as $\mathcal{N}^\perp \subseteq \ker(f)$ we have $f \in (\mathcal{N}^\perp)^0$ so $\mathcal{N} \subseteq (\mathcal{N}^\perp)^0$.

c. We see that by definition $\alpha = \pi^\dagger \in L((X/\mathcal{M})^*, X^*)$ and it is injective as $\pi$ is surjective (and hence has its range is obviously dense in $X/\mathcal{M}$) so its an isomorphism on its range. If $x \in \mathcal{M}$ then $\alpha(f)(x) = f(\pi(x)) = f(0 + \mathcal{M}) = 0$ so $\alpha$'s range is a subset of $\mathcal{M}^0$ and if $f \in \mathcal{M}^0$ then just as in Exercise 5.15 it induces an unique $S \in (X/\mathcal{M})^*$ as $S(x + \mathcal{M}) = f(x)$ and thus $S \circ \pi = f$ (continuity follows with the same arguement as in the linked problem) showing surjectivity. So we have proved that this is an isomorphism onto $\mathcal{M}^0$. To show that this is an isometry it suffices to notice that we can follow the exact steps (the closed subspace $\mathcal{M}$ replaces $\mathcal{N}(T)$ there) as in the linked problem to show that $\|f \circ \pi\| = \|f\|$ for all such $f \in (X/\mathcal{M})^*$.

d. As $\ker \beta = \mathcal{M}^0$ we have an induced continuous linear map $\overline{\beta} : X^*/\mathcal{M}^0 \to \mathcal{M}^*$ as in Exercise 5.15. Given any $g \in \mathcal{M}^*$ we can use hahn banach with the seminorm $x \mapsto \|g\|\|x\|$ to get $f \in X^*$ such that $f|_\mathcal{M} = g$ and thus, $\beta(f) = g$ so $\beta$ is surjective and thus so is $\overline{\beta}$ by construction. Here, $\|\overline{\beta}(f + \mathcal{M}^0)\| = \|f|_\mathcal{M}\|$ and we claim that $\|f|_\mathcal{M}\| = \|f + \mathcal{M}^0\|$. If $g \in \mathcal{M}^0$ then $\|f|_\mathcal{M}\| = \|(f + g)|_\mathcal{M}\| \leq \|f + g\|$ where the second inequality holds as in the last term the supremum is taken over a larger set and thus $\|f|_\mathcal{M}\| \leq \|f + \mathcal{M}^0\|$. Now, as we did to show surjectivity we can construct $F \in X^*$ with $F|_\mathcal{M} = f|_\mathcal{M}$ such that $|F(x)| \leq \|f_\mathcal{M}\|\|x\|$ so $\|F\| \leq \|f_\mathcal{M}|$ and as $F - f \in \mathcal{M}^0$ by construction, we have $\|f + \mathcal{M}^0\| \leq \|f + (F - f)\| \leq \|f|_\mathcal{M}\|$ so it is indeed an isometry. As its an isometry, its injective and thus its a bijection all in all as well as being an isomorphism due to isometry again.

□

**Exercise 5.24.** Suppose that $X$ is a Banach space.

a. Let $\widehat{X}$, $\widehat{(X^*)}$ be the natural images of $X$, $X^*$ in $X^{**}$, $X^{***}$, and let $\widehat{X}^0 = \{F \in X^{***} : F|_{\widehat{X}} = 0\}$. Then $\widehat{(X^*)} \cap \widehat{X}^0 = \{0\}$ and $\widehat{(X^*)} + \widehat{X}^0 = X^{***}$.

b. $X$ is reflexive iff $X^*$ is reflexive.

*Solution.*

a. If $F \in \widehat{X}^0 \cap \widehat{(X^*)}$, say $F = \hat{f}$ for $f \in X^*$, then for all $\hat{x} \in \widehat{X}$, $\hat{f}(\hat{x}) = \hat{x}(f) = f(x) = 0$ so $f = 0$ and we have $\hat{f} = F = 0$ so $\widehat{(X^*)} \cap \widehat{X}^0 = \{0\}$. Now, for $\psi \in X^{***}$ consider $\phi = \psi|_{\widehat{X}}$, here we see that $\hat{x} \mapsto \phi(\hat{x})$ corresponds to $\phi_* \in X^*$ defined by $\phi_*(x) := \phi(\hat{x})$ and we find that $\phi = \widehat{\phi_*}|_{\widehat{X}}$ as $\widehat{\phi_*}(\hat{x}) = \hat{x}(\phi_*) = \phi_*(x) = \phi(\hat{x})$. It is clear that $\psi - \widehat{\phi_*} \in \widehat{X}^0$ so we are done.

b. If $X$ is reflexive then $\widehat{X} = X^{**}$ so $\widehat{X}^0 = (X^{**})^0 = \{0\}$ and by part (a), $\widehat{(X^*)} = X^{***}$ so $X^*$ is reflexive. Conversely if $X^*$ is reflexive then $\widehat{(X^*)} = X^{***}$ and by part (a) this forcrs $\widehat{X}^0 = \{0\}$. As $X$ is a banach space, $\widehat{X}$ is a closed subset of $X^{**}$ and if it was a strict subset then by Theorem 5.8a we could find a nonzero functional that was zero on it, but this is not the case as $\widehat{X}^0 = \{0\}$ so it must be all of $X^{**}$ proving reflexivity for $X$ and we are done.

□

**Exercise 5.25.** If $X$ is a Banach space and $X^*$ is separable, then $X$ is separable. (Let $\{f_n\}_1^\infty$ be a countable dense subset of $X^*$. For each $n$ choose $x_n \in X$ with $\|x_n\| = 1$ and $|f_n(x_n)| \geq \frac{1}{2}\|f_n\|$. Then the linear combinations of $\{x_n\}_1^\infty$ are dense in $X$.) Note: Separability of $X$ does not imply separability of $X^*$.

*Solution.* $\mathbb{K}$ always has a seperable subfield ($\mathbb{Q}$ or $\mathbb{Q} + i\mathbb{Q}$) and it can be easily shown that the closure of the linear combinations of $\{x_n\}_1^\infty$ over $\mathbb{K}$ is the same as that over this seperable subfield and as the span of countably many vectors over a countable field is countable (this follows via basic set theoretic arguements) we would be done if we prove the claim in the hint. If the closure of the span was a strict subspace (it is a subspace by Exercise 5.5) then we could find $f \in X^*$ such that $f \neq 0$ but $f$ is zero on the closure. If $\|f - f_n\| \leq \epsilon$ then $\epsilon > |(f_n - f)(x_n)| = |f_n(x_n)| \geq \|f_n\|/2$ and thus, $\|f\| \leq \|f_n\| + \|f - f_n\| \leq 2\epsilon + \epsilon$. By density, this contraidcts the fact that $f \neq 0$ so we get a contradiction and hence the span was dense to begin with. □

**Exercise 5.26.** Let $X$ be a real vector space and let $P$ be a subset of $X$ such that (i) if $x, y \in P$, then $x + y \in P$, (ii) if $x \in P$ and $\lambda \geq 0$, then $\lambda x \in P$ (iii) if $x \in P$ and $-x \in P$, then $x = 0$. (Example: If $X$ is a space of real-valued functions, $P$ can be the set of nonnegative functions in $X$.)

a. The relation $\leq$ defined by $x \leq y$ iff $y - x \in P$ is a partial ordering on $X$.

b. (The Krein Extension Theorem) Suppose that $\mathcal{M}$ is a subspace of $X$ such that for each $x \in X$ there exists $y \in \mathcal{M}$ with $x \leq y$. If $f$ is a linear functional on $\mathcal{M}$ such that $f(x) \geq 0$ for $x \in \mathcal{M} \cap P$, there is a linear functional $F$ on $X$ such that $F(x) \geq 0$ for $x \in P$ and $F|_{\mathcal{M}} = f$. (Consider $p(x) = \inf\{f(y) : y \in \mathcal{M} \text{ and } x \leq y\}$.)

*Solution.*

a. $P$ is what we usually call a *cone*, this is trivially a partial order.

b. For $x, y \in X$ suppose $x_n, y_n \in \mathcal{M}$ such that $x_n \geq x, y_n \geq y$ and $f(x_n) \to p(x), f(y_n) \to p(y)$ then $x_n + y_n \geq x + y$ as $(x_n - x), (y_n - y) \in P$ and $x_n + y_n \in \mathcal{M}$ thus $p(x+y) \leq f(x_n+y_n) = f(x_n)+f(y_n) \to p(x)+p(y)$ and clearly $p(\lambda x) = \lambda p(x)$ for $\lambda \geq 0$ stemming from the fact that $x \in P \iff \lambda x \in P$ for $\lambda > 0$. This shows $p$ to be a sublinear functional. For $x \in \mathcal{M}$ and $y \in \mathcal{M}$ such that $x \leq y$ we have $f(x) = f(y) - f(y-x)$ where $y - x \in P \cap \mathcal{M}$ so $f(x) \leq f(y)$ as $f(y-x) \geq 0$ proving $f(x) \leq p(x)$. Now, we can use hanh banach to extend $f$ to $F$ such that $F|_{\mathcal{M}} = f$ and $F \leq p$. If $x \in P$ then we have $F(x) = -F(-x) \geq -p(-x)$ so it suffices to show that $p(-x) \leq 0$. As $0 - (-x) = x \in P$ we have $-x \leq 0$ and as $0 \in \mathcal{M}$, by definition $p(-x) \leq f(0) = 0$ so we are done.

□

**Exercise 5.27.** There exist meager subsets of $\mathbf{R}$ whose complements have Lebesgue measure zero.

*Solution.* We can find fat cantor sets $K_n \subseteq [0, 1]$ that are dense nowhere but have

lebesgue measure atleast $1 - 1/n$, then consider

$$K = \bigcup_{n\in\mathbb{Z}} \left( n + \bigcup_{m\in\mathbb{N}} K_m \right)$$

This is clearly meager and for all $[n, n+1]$ we have $\mu(K \cap [n, n+1]) \geq \mu(\cup_{m\in\mathbb{N}} K_m) \geq 1 - 1/m$ for all $m$ implying $\mu(K \cap [n, n+1]) = 1$ so, $\mu(K^c \cap [n, n+1]) = 0$ and $\mu(K^c) = \sum_1^\infty \mu(K^c \cap [n, n+1]) = 0$ provivng the claim. □

**Exercise 5.28.** The Baire category theorem remains true if $X$ is assumed to be an LCH space rather than a complete metric space. (The proof is similar; the substitute for completeness is Proposition 4.21.)

*Solution.* For any $x \in X$, any open nbhd $W$ of $x$ will have nonempty intersection with $U_1$ and by proposition 4.30 we can choose a compact neighbourhood $K_1$ of $y_1$ where $y_1 \in U_1 \cap W$ such that $K_1 \subseteq W \cap U_1$. As $K_1$ is a nbhd we can find an open nbhd $H$ of $y_2$ in $K_1$ and repeating the above arguement with $H$ replacing $W$ and $U_2$ replacing $U_1$ we can find a compact nbhd $K_2$ of $y_2$ such that $K_2 \subseteq H \cap U_2 \subseteq K_1 \subseteq U_1$ and we keep doing this to inductively construct a decreasing sequence of nonempty compact sets $\{K_n\}$ such that $K_n \subseteq U_n$. Now, by proposition 4.21 (finite intersection property), as $K_1$ is compact and contains $K_n$ for all $n$, which in turn are also relatively compact in $K_1$ (this has been proved in the beginning of Exercise 4.49), we can say that $\cap_n K_n$ is nonempty and for any $p$ as such, it is contained in all $U_n$ and $p \in W$ and as $W$ was arbitrary we can say that $\cap U_n$ is dense. For the other part it suffices to notice that if $F_n$ are nowhere dense sets then $(\overline{F_n})^c$ is dense and open so $\cap(\overline{F_n})^c$ is nonempty by the previous proof, so $\cup F_n \subseteq \cup \overline{F_n} = (\cap(\overline{F_n})^c)^c$ is not all of $X$. □

**Exercise 5.29.** Let $Y = L^1(\mu)$ where $\mu$ is counting measure on $\mathbb{N}$, and let $X = \{f \in Y : \sum_1^\infty n|f(n)| < \infty\}$, equipped with the $L^1$ norm.

a. $X$ is a proper dense subspace of $Y$; hence $X$ is not complete.

b. Define $T : X \to Y$ by $Tf(n) = nf(n)$. Then $T$ is closed but not bounded.

c. Let $S = T^{-1}$. Then $S : Y \to X$ is bounded and surjective but not open.

*Solution.*

a. $f(n) = n^{-2}$ is in $Y$ but not $X$ so its indeed a proper subspace (the fact that it is a subspace is trivially shown.) If $f \in Y$, for any $\epsilon > 0$ we can find $N$ such that $\|f\chi_{(N,\infty)}\| < \epsilon$, thus $f\chi_{[0,N]} \in X$ and $\|f - f\chi_{[0,N]}\| = \|f\chi_{(N,\infty)}\| < \epsilon$ proving density. As complete sets are also closed we see that this can't be complete as otherwise it would be all of $Y$ due to density, which we showed to be false at the start.

b. Say $(f_m, Tf_m) \in \Gamma(T)$ converges in $X \times Y$ to say $(f, g)$, then as $f_m \to f$ in $L^1$ we have pointwise convergence as for all $n, |f_m(n) - f(n)| \leq \|f_m - f\| \to 0$ and similary for $Tf_m$, which converging pointwise implies $Tf_m \to Tf = g$. We also see that as $Tf_m \to Tf$ in $L^1$ with $Tf_m$ being $L^1, Tf$ is also $L^1$ so $f \in X$ (notice that $X = \{f \in Y : Tf \in Y\}$)and thus $(f, g) = (f, Tf) \in \Gamma(T)$ proving that it is closed. This is however not bounded because $\|T\chi_{\{n\}}\| = n = n \cdot \|\chi_{\{n\}}\|$ for all $n$.

c. Firstly, $T$ is clearly invertible (injectivity is clear and surjectivity follows from the definition of $Y$ easily.) $S$ being open would make $T$ continuous (equivalently, bounded) so it can't be, by the previous part. $S$ maps $f$ to $f \cdot (\mathrm{Id})^{-1}$ and thus $|Sf| \leq |f|$ implying $\|Sf\| \leq \|f\|$ so its bounded. We have surjectivity as its a bijection already.

□

**Exercise 5.30.** Let $Y = C([0,1])$ and $X = C^1([0,1])$, both equipped with the uniform norm.

a. $X$ is not complete.

b. The map $(d/dx) : X \to Y$ is closed (see Exercise 9) but not bounded.

*Solution.*

a. Non differentiable functions that are continuous on $[0,1]$ exist so $X$ is not complete by the stone-weirstrass approximation theorem (polynomials are differentiable.)

b. It is not bounded as $(d/dx)\exp(nx) = n \cdot \exp(nx)$. This being closed follows directly from Exercise 5.9 as $(f_n, f_n')$ converging in the product norm to $(f, g)$ implies pointwise convergence in the uniform norm too.

□

**Exercise 5.31.** Let $X, Y$ be Banach spaces and let $S : X \to Y$ be an unbounded linear map (for the existence of which, see §5.6). Let $\Gamma(S)$ be the graph of $S$, a subspace of $X \times Y$.

a. $\Gamma(S)$ is not complete.

b. Define $T : X \to \Gamma(S)$ by $Tx = (x, Sx)$. Then $T$ is closed but not bounded.

c. $T^{-1} : \Gamma(S) \to X$ is bounded and surjective but not open.

*Solution.*

a. Complete sets are closed and thus if this was complete then $T$ would be bounded by the closed graph theorem, thus is not complete.

b. If $\|Sx\| \geq n\|x\|$ for $n \in \mathbb{N}$ then $\|(Sx, x)\| = \|Sx\| \geq n\|x\|$ as $\|Sx\| \geq \|x\|$ so this is unbounded too as such $x$ exist given unboundedness of $S$. The maps $\pi_1 : X \times \Gamma(S) \to X$ defined by $(x, (t, S(t)) \mapsto x$ and $\pi_2$with the same domain and codomain mapping $(x, (t, S(t))$ to $t$ are both clearly continuous and as $X$ is hausdorff (being a metric space), by Exercise 4.16 we see that $\Gamma(T) = \{p \in X \times \Gamma(S) : \pi_1(p) = \pi_2(p)\}$ is closed (this can also be done by considering $\ker(\pi_1 - \pi_2)$ with a bit more justification.)

c. This maps $(x, Sx)$ to $x$ and as $\|(x, Sx)\| \geq \|x\|$, it is bounded. It is clearly surjective and if it was open then the open mapping theorem would imply boundedness of $T$ which is false.

□

**Exercise 5.32.** Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on the vector space $X$ such that $\|\cdot\|_1 \leq \|\cdot\|_2$. If $X$ is complete with respect to both norms, then the norms are equivalent.

*Solution.* Consider the identity linear map from $X$ with norm $\|\cdot\|_2$ to $X$ with norm $\|\cdot\|_1$, this is clearly bounded by hypothesis (infact $\|\cdot\|_1 \leq C\|\cdot\|_2$ would do as well) and by corollary 5.11, it's inverse is an isomorphism so we can find $K > 0$ with $\|\cdot\|_1 \geq K\|\cdot\|_2$ and we are done as equivalent norms generate the topologies. □

**Exercise 5.33.** There is no slowest rate of decay of the terms of an absolutely convergent series; that is, there is no sequence $\{a_n\}$ of positive numbers such that $\sum a_n|c_n| < \infty$ iff $\{c_n\}$ is bounded. (The set of bounded sequences is the space $B(\mathbb{N})$ of bounded functions on $\mathbb{N}$, and the set of absolutely summable sequences is $L^1(\mu)$ where $\mu$ is counting measure on $\mathbb{N}$. If such an $\{a_n\}$ exists, consider $T : B(\mathbb{N}) \to L^1(\mu)$ defined by $Tf(n) = a_n f(n)$. The set of $f$ such that $f(n) = 0$ for all but finitely many $n$ is dense in $L^1(\mu)$ but not in $B(\mathbb{N})$.)

*Solution.* Consider $B(\mathbb{N})$ with the uniform norm. If such a sequence did exist, then following the hint, $T$ is well defined. We also clearly see that $\|Tf\| \leq \|f\|\|T\mathbf{1}\|$ where $\mathbf{1} \in B(\mathbb{N})$ is identically 1, proving continuity. As $a_n \neq 0$ this is an injection and for the same reason this is also a surjection (it follows from the fact that $L^1(\mathbb{N}, \mu) \subseteq B(\mathbb{N})$) so by the open mapping theorem(Corollary 5.11 to be precise) we must have $T^{-1}$ be bounded. Let $C_0(\mathbb{N})$ be the set of functions that are zero on all but finitely many values. From the hint, with the norms in mind we see that $C_0(\mathbb{N})$ is dense in $L^1(\mu)$ but not in $B(\mathbb{N})$ but as $T^{-1}$ is a is a continuous bijectiion we must have $T^{-1}(C_0(\mathbb{N}))$ dense in $B(\mathbb{N})$ but this is false as $T^{-1}$ only changes functions at each number by a fixed nonzero multiple so $T^{-1}(C_0(\mathbb{N})) = C_0(\mathbb{N})$ and we have a contradiction. □

**Exercise 5.34.** With reference to Exercises 9 and 10, show that the inclusion map of $L^1_k([0,1])$ into $C^{k-1}([0,1])$ is continuous (a) by using the closed graph theorem, and (b) by direct calculation. (This is to illustrate the use of the closed graph theorem as a labor-saving device.)

*Solution.*

a. If $(f_n, f_n)$ converges to $(f, g)$ in $L^1_k \times C^{k-1}$ then we want to show that $f = g$, after which closed graph theorem implies continuity. The components must converge too so we always have, by the definition of these norms, $f_n \to f$ in $L^1$ and $f_n \to g$ uniformly so $f_n \to g$ in $L^1$ too and thus $f = g$ a.e but as $f, g$ are continuous by inclusion in $C^{k-1} \supseteq L^1_k$, we have $f = g$.

b. This time we want to show that $\|f\|_{C^{k-1}} \leq C\|f\|_{L^1_k}$ for some $C > 0$. For $j = 1, \ldots, k$, as $f^{(j-1)}$ is continuous, so is $|f^{(j-1)}|$ and we can find some $x_{j-1}$ such that $|f^{(j-1)}(x_{j-1})| = \int_0^1 |f^{(j-1)}(t)|\,\mathrm{d}t$ and thus, for all $x$, by the fundamental theorem of calculus (for $j = k$ we use absolute continuity)

$$|f^{(j-1)}(x)| = \left| f^{(j-1)}(x_{j-1}) + \int_{x_{j-1}}^{x} f^{(j)}(t)\,\mathrm{d}t \right| \leq \int_0^1 |f^{(j-1)}(t)|\,\mathrm{d}t + \int_0^1 |f^{(j)}(t)|\,\mathrm{d}t$$

Thus, we have

$$\|f\|_{C^{k-1}} = \sum_{j=1}^{k} \|f^{(j-1)}\|_u \leq \sum_{j=1}^{k} \left( \int_0^1 |f^{(j-1)}(t)|\,\mathrm{d}t + \int_0^1 |f^{(j)}(t)|\,\mathrm{d}t \right) \leq 2\|f\|_{L^1_k}$$

So we are done. We can confidently say that the closed graph theorem makes life easier.

□

**Exercise 5.35.** Let $X$ and $Y$ be Banach spaces, $T \in L(X,Y)$, $\mathcal{N}(T) = \{x : Tx = 0\}$, and $\mathcal{M} = \text{range}(T)$. Then $X/\mathcal{N}(T)$ is isomorphic to $\mathcal{M}$ iff $\mathcal{M}$ is closed. (See Exercise 15.)

*Solution.* If $\mathcal{M}$ is closed then it is complete as $Y$ is complete and by Exercise 5.12d $X/\mathcal{N}(T)$ is complete as the kernel is closed. Following the notation of Exercise 5.15 we see that $T$ induces a map $S \in L(X/\mathcal{N}(T), Y)$, if we look at this as a continuous linear map into $\mathcal{M}$ then this is clearly bijective and by Corollary 5.11 it is an isomorphism. For the converse it suffices to prove that if $H$ is a banach space isomorphic to a normed vector space $K$ then $K$ is also a banach space. Indeed if these are isomorphic through $\phi$, for $k_n \in K$ such that $\sum_1^\infty \|k_n\| < \infty$ we see that $\sum_1^\infty \|\phi^{-1}k_n\| \leq \|\phi^{-1}\| \sum_1^\infty \|h_n\| < \infty$ so $\sum_1^\infty \phi^{-1}k_n$ converges, which implies, by continuity of $\phi$, convergence of $\sum_1^\infty k_n$ so by Theorem 5.1 we are done. □

**Exercise 5.36.** Let $X$ be a separable Banach space and let $\mu$ be counting measure on $\mathbb{N}$. Suppose that $\{x_n\}_1^\infty$ is a countable dense subset of the unit ball of $X$, and define $T : L^1(\mu) \to X$ by $Tf = \sum_1^\infty f(n)x_n$.

a. $T$ is bounded.

b. $T$ is surjective.

c. $X$ is isomorphic to a quotient space of $L^1(\mu)$. (Use Exercise 35.)

*Solution.*

a. Linearity is trivially shown and as $x_n$ are in the unit ball, $\|Tf\| \leq \sum_1^\infty |f(n)|\|x_n\| \leq \|f\|_1$ proving boundedness

b. For any $x \neq 0$ we can find $x_{n_1}$ such that $\|\|x\| \cdot x_{n_1} - x\| \leq 1/2$ by hypothesis and again we can find $x_{n_2}$ (it is possible to choose $n_2 \neq n_1$ as omitting finitelyelements does not affect density) with $0 \leq \alpha_2 \leq 1/2$ such that $\|\alpha x_{n_2} + \|x\| \cdot x_{n_1} - x\| \leq 1/4$, we can keep going to have a sequence $\{n_k\} \subseteq \mathbb{N}$ with unique terms and $\{\alpha_k\} \subseteq \mathbb{R}$ with $\alpha_1 = \|x\|$ and $0 \leq \alpha_j \leq 2^{1-j}$ for $j \geq 2$ such that $\|x - \sum_1^j \alpha_j x_{n_j}\| \leq 2^{-j}$ and thus $\sum_1^\infty \alpha_j x_{n_j} = x$. Now, let $f$ be defined as $f(n_j) = \alpha_j$ and 0 otherwise, then it is clearly $L^1$ and $\sum_1^\infty f(n)x_n = x$ proving surjectivity.

c. By Theorem 2.25, if $\sum_1^\infty \|f_n\| < \infty$ then $\sum f_n$ converges a.e. (here, this implies convergence everywhere as we work with a counting measure) to a function in $L^1(\mu)$ proving completeness of $L^1(\mu)$ by theorem 5.1. Now, as $T$ is surjective as shown above, by Exercise 5.35 we must have $X$ isomorphic to $L^1(\mu)/\mathcal{N}(T)$.

□

**Exercise 5.37.** Let $X$ and $Y$ be Banach spaces. If $T : X \to Y$ is a linear map such that $f \circ T \in X^*$ for every $f \in Y^*$, then $T$ is bounded.

*Solution.* By the closed graph theorem, to prove continuity it suffices to show that if $x_n \to x$ and $Tx_n \to y$ then $y = Tx$. By hypothesis we have $(f \circ T)(x_n) \to (f \circ T)(x)$ and $f(Tx_n) \to f(y)$ thus $f(y) = f(Tx)$ for all $f \in Y^*$. By theorem 5.8c we know that bounded linear functionals seperate points and thus we must have $y = Tx$ so we are done. □

**Exercise 5.38.** Let $X$ and $Y$ be Banach spaces, and let $\{T_n\}$ be a sequence in $L(X, Y)$ such that $\lim T_n x$ exists for every $x \in X$. Let $Tx = \lim T_n x$; then $T \in L(X, Y)$.

*Solution.* By the uniform bounded principle, as $T_n x$ converges for each $x$, this sequence must be bounded and we have $\sup_{n\in\mathbb{N}} \|T_n x\| < \infty$ so by the uniform boundedness princple $\sup_{n\in\mathbb{N}} \|T_n\| < \infty$ and if we let this be $C$ then $\|Tx\| = \|\lim T_n x\| = \lim \|T_n x\| \leq \lim \|T_n\|\|x\| \leq C\|x\|$ by continuity of the norm so we are done. □

**Exercise 5.39.** Let $X, Y, Z$ be Banach spaces and let $B : X \times Y \to Z$ be a separately continuous bilinear map; that is, $B(x, \cdot) \in L(Y, Z)$ for each $x \in X$ and $B(\cdot, y) \in L(X, Z)$ for each $y \in Y$. Then $B$ is jointly continuous, that is, continuous from $X \times Y$ to $Z$. (Reduce the problem to proving that $\|B(x, y)\| \leq C\|x\|\|y\|$ for some $C > 0$.)

*Solution.* As $\sup_{\|x\|\leq 1} \|B(x, y)\| = \|B(\cdot, y)\| < \infty$ for all $y$, by the uniform boundedness principle we have $\sup_{\|x\|\leq 1} \|B(x, \cdot)\| < \infty$. This implies that the linear map $x \mapsto B(x, \cdot)$ is bounded and thus we can find $C$ with $\|B(x, \cdot)\| \leq C\|x\|$ so

$$\|B(x, y)\| \leq \|B(x, \cdot)\|\|y\| \leq C\|x\|\|y\|.$$

Continuity follows by the sequential criterion: if $(x_n, y_n) \to (x, y)$ then

$$\begin{aligned}\|B(x_n, y_n) - B(x, y)\| &\leq \|B(x_n, y_n) - B(x_n, y)\| + \|B(x_n, y) - B(x, y)\| \\ &\leq C(\|x_n\|\|y_n - y\| + \|x_n - x\|\|y\|) \longrightarrow 0\end{aligned}$$

so we are done. □

**Exercise 5.40.** (The Principle of Condensation of Singularities) Let $X$ and $Y$ be Banach spaces and $\{T_{jk} : j, k \in \mathbb{N}\} \subset L(X, Y)$. Suppose that for each $k$ there exists $x \in X$ such that $\sup\{\|T_{jk}x\| : j \in \mathbb{N}\} = \infty$. Then there is an $x$ (indeed, a residual set of $x$'s) such that $\sup\{\|T_{jk}x\| : j \in \mathbb{N}\} = \infty$ for all $k$.

*Solution.* We will show that its complement

$$\bigcup_{k\in\mathbb{N}} \{x \in X : \sup_{j\in\mathbb{N}} \|T_{jk}x\| < \infty\}$$

is meager. If $\{x \in X : \sup_{j\in\mathbb{N}} \|T_{jk}x\| < \infty\}$ is non meager then by the uniform boundedness principle we must have $\sup_{j\in\mathbb{N}} \|T_{jk}\| < \infty$ which implies that for all $x \in X$

$$\|T_{jk}x\| \leq \|x\| \sup_{j\in\mathbb{N}} \|T_{jk}\| < \infty \quad \text{for all } j \in \mathbb{N}$$

which is a contradiction so these must be meager, let it be $\cup_{j\in\mathbb{N}} E_j^{(k)}$ with $E_j^{(k)}$ all nowhere dense. Then,

$$\bigcup_{k\in\mathbb{N}} \{x \in X : \sup_{j\in\mathbb{N}} \|T_{jk}x\| < \infty\} = \bigcup_{k\in\mathbb{N}} \bigcup_{j\in\mathbb{N}} E_j^{(k)} = \bigcup_{(k,j)\in\mathbb{N}^2} E_j^{(k)}$$

which a countable union of nowhere dense sets, and thus its complement (which is the setwe care about) is residual. □

**Exercise 5.41.** Let $X$ be a vector space of countably infinite dimension (that is, every element is a finite linear combination of members of a countably infinite linearly independent set). There is no norm on $X$ with respect to which $X$ is complete. (Given a norm on $X$, apply Exercise 18b and the Baire category theorem.)

*Solution.* Say $\{x_n\}_1^\infty$ is a basis, let $E_k := \text{Span}\{x_j : 1 \leq j \leq k\}$, then $\cup E_k = X$. If we manage to show that $E_k$ are all nowhere dense then we would have a contradictiondue to BCT and prove the claim. By Exercise 5.18b $E_k$ is closed for all $k$, for the sake of contradiction say $E_k$ is not nowhere dense and we can find some $x$ and $r > 0$ with $B(x, r) \subseteq \overline{E_k} = E_k$ but then, as $E_k$ is closed under dilation we must have $E_k = X$ which contradicts infinite dimensionality of $X$ so we are done. □

**Exercise 5.42.** Let $E_n$ be the set of all $f \in C([0, 1])$ for which there exists $x_0 \in [0, 1]$ (depending on $f$) such that $|f(x) - f(x_0)| \leq n|x - x_0|$ for all $x \in [0, 1]$.

a. $E_n$ is nowhere dense in $C([0, 1])$. (Any real $f \in C([0, 1])$ can be uniformly approximated by a piecewise linear function $g$ whose linear pieces, finite in number, have slope $\pm 2n$. If $\|h - g\|_u$ is sufficiently small, then $h \notin E_n$.)

b. The set of nowhere differentiable functions is residual in $C([0, 1])$.

*Solution.*

a. If $f \in C([0, 1])$ then it is uniformly continuous and bounded on $[0, 1]$ from which we can easily construct such $g$ as in the hint by making the the intervals on which it is linear small enough (first approximate it with a piecewise linear function and then approximate this by functions in $g_n$.) Fix $g \in E_n$, and let the intervals on which its linear have length atleast $L > 0$ (we can do this as there are finitely many such intervals by definition.) For all $x_0 \in [0, 1]$ we can look at the interval its in and conclude that atleast one of the endpoints of this interval is atleast $L/2$ away from $x_0$, let it be $x$ (thus $|x - x_0| \geq L/2$). If we have $\|h - g\|_u < L/4$ then by construction we must have $|h(x) - h(x_0)| > |g(x) - g(x_0)| - 2(L/4) \geq 2n|x - x_0| - |x - x_0| \geq n|x - x_0|$ and as $x_0$ was arbitrary we must have $h \notin E_n$. For any $f \in C([0, 1])$ and $\epsilon > 0$ we can find such a $g \in B(f, \epsilon)$ and there will exist $\epsilon_* > 0$ such that $B(g, \epsilon_*) \subseteq E_n^c$ so $g \notin \overline{E_n}$ implying $B(f, \epsilon) \subsetneq (\overline{E_n})^\circ$ and thus $E_n$ is nowhere dense in $C([0, 1])$.

b. The complement of the set of nowhewre differntiable functions (call it $S$) contains functions that are differentiable somewhere, say $x_0$ then we can find some $\delta > 0$ such that

$$\sup_{x \in [0,1] \cap B(x_0, \delta)} \left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq 1$$

as the limit of this expression as $x \to x_0$ (from inside $[0, 1]$) exists and is thus bounded near $x_0$. We also see that $k(x) := |f(x) - f(x_0)|/|x - x_0|$ is a continuous function on the compact set $[0, 1] \setminus B(x_0, \delta)$ and has a supremum, let it be $M$. Then, by construction $f \in E_n$ if $n \geq \max\{M, 1\}$. Thus, $S^c \subseteq \cup E_n$ and we can write $S^c = \cup(E_n \cap S^c)$ and as $E_n \cap S$ are nowhere dense too, $S^c$ is meager and $S$ residual so we are done.

□

**Exercise 5.43.** Prove Proposition 5.16. (For part (b), proceed as in Exercise 56d in §4.5.)

*Solution.*

a. If $x \neq y$ and $p_\alpha(x-y) > 0$ (let this be $2\epsilon$) then $U_{x\alpha\epsilon}, U_{y\alpha\epsilon}$ are open sets in this topology that contain $x, y$ respectively and they are disjoint for if $z$ is in both then $p_\alpha(x-y) \leq p_\alpha(x-z)+p_\alpha(z-y) < \epsilon+\epsilon$ which is not possible, making our topology hausdorff. For the converse if $x \neq 0$ is such that $p_\alpha(x) = 0$ for all $\alpha$ then from proposition 5.14, for any open sets $U$ containing 0 we can find $V = \cap_1^k U_{0\alpha_k\epsilon_k}$ such that $0 \in V \subseteq U$ and by hypothesis $x \in U_{0\alpha\epsilon}$ for all $\alpha$ and $\epsilon > 0$ implying $x \in U_{0\alpha_k\epsilon_k}$ for all $k$ so $x \in U$ too and we fail to seperate 0 from $x$ with open sets, making the topology non hausdorff.

b. Wlog we let $A = \mathbb{N}$, then consider

$$\rho(x,y) := \sum_{k=1}^{\mathbb{N}} \frac{1}{2^k} \cdot \frac{p_k(x-y)}{1+p_k(x-y)}$$

The triangle inequality is satisfied and the proof in Exercise 4.56 can be repurposed here. It is symmetric and from part (a) we have that $\rho(x,y) = 0$ iff $x = y$, proving that this is a metric and translation invariance follows from that of the $p_k$.
We use the following lemma: If $X$ is a topological space with two topologies $E, F$ then $E = F$ whenever convergence of nets in either is equivalent i.e. $x_\alpha \to x$ in $E$ iff $x_\alpha \to x$ in $F$. By symmetry it suffices to show that if $x_\alpha \to x$ in $E$ implies that in $F$ then $E \supseteq F$. Consider the identity map $i$ from $(X,E)$ to $(X,F)$. As $i(x_\alpha) \to i(x)$ in $F$ whenever $x_\alpha \to x$ making $i$ continuous and $U \in F \implies i^{-1}(U) = U \in E$ hence proving the lemma.
If $x_\alpha \to x$ in the topology generated by those norms then for all $\epsilon > 0$ and $n \in \mathbb{N}$ we can find $\beta$ such that $\alpha \gtrsim \beta \implies p_j(x_\alpha - x) < \epsilon$ for all $j \leq n$ (find a $\beta$ for each $j$ and take some element greater than all of them, we can do this by the definition of directed sets which nets are indexed with.) Then,

$$\rho(x_\alpha, x) \leq \sum_{1\leq j\leq n} \frac{\epsilon}{2^j(1+\epsilon)} + \sum_{j>n} \frac{1}{2^j} \leq \epsilon + 2^{-n}$$

Now, if we choose $n$ large enough and $\epsilon$ small enough, for all $r > 0$ we can find $\beta$ such that $\alpha \gtrsim \beta \implies x_\alpha \in B(x,r)$ and we thus have convergence in the metric. If we begin with convergence in the metric then the fact that $\rho(x,y) \geq 2^{-k}p_k(x-y)/(1+p_k(x-y))$ for all $k$, implies that $\rho(x_\alpha, x) \to 0$ using properties of $\Phi$ as in Exercise 4.56 (which is equivalent to the net converging in the metric, this is easily shown) implies $p_k(x_\alpha - x) \to 0$ for all $k$ and we thus have convergence in the topology generated by $\{p_k\}_{k\in\mathbb{N}}$ too by proposition 5.14b and we can thus conclude by the lemma.

□

**Exercise 5.44.** If $X$ is a first countable topological vector space and every Cauchy sequence in $X$ converges, then every Cauchy net in $X$ converges.

*Solution.* It suffices to prove that convergence of cauchy sequences forces convergence of nets. Suppose $\langle x_i \rangle_{i\in I}$ is a cauchy net, i.e. $\langle x_i - x_j \rangle_{(i,j)\in I\times I}$ converges to 0. Because the topology is first countable we can find a countable decreasing neighborhood base of 0, $\{V_n\}_{n\in\mathbb{N}}$. We can find $\alpha_n$ inductively such tha that $\beta,\gamma \geq \alpha_n \implies x_\beta - x_\gamma \in V_n$ and $\alpha_k \gtrsim \alpha_{k-1}$. The sequence $(x_{\alpha_k})_{k\in\mathbb{N}}$ is clearly cauchy too (in the net sense) as $\left\langle x_{\alpha_i} - x_{\alpha_j} \right\rangle_{(i,j)\in\mathbb{N}\times\mathbb{N}}$ is a cauchy net as $m,n \geq k \implies \alpha_m, \alpha_n \geq \alpha_k$ and thus $x_{\alpha_m} - x_{\alpha_n} \in V_k$ so we have cauchy convergence as $V_j$ formed a decreasing countable neighbourhood base. Now, as all cauchy sequences converge, we must have some $x$ which $(x_{\alpha_k})$ coverges to. As this is a TVS, addition is continuous so given any open $U$ containing $x$, as $x = x + 0 \in U$ we can find some open (in the product topology) $E \times F$ such that $x \in E, 0 \in F$ and $E + F \subseteq U$ implying $x + F \subseteq U$. By construction, for all $k$, we can find $m_k$ such that $x - x_{\alpha_{m_k}} \in V_k$ and then we can find open neighborhoods $F_k$ of 0 such that $(x - x_{\alpha_{m_k}}) + F_k \subseteq V_k$ and we can also find, by definition of cauchy nets, $\beta$ such that $\beta_1, \beta_2 \gtrsim \beta$ implies $x_{\beta_1} - x_{\beta_2} \in F_k$ and now for any $\gamma \gtrsim \beta, \alpha_{m_k}$ we can write

$$x - x_\zeta = (x - x_{\alpha_{m_k}}) + (x_{\alpha_{m_k}} - x_\zeta) \in (x - x_{\alpha_{m_k}}) + F_k \subseteq V_k \quad \text{for all } \zeta \geq \gamma$$

proving that $x_\alpha \to x$ as $V_k$ formed a countable decreasing neighborhood base so this net converges. This is a mimicry of the proof of cauchy sequences with convergent subsequences converging but at higher generality. □

**Exercise 5.45.** The space $C^\infty(\mathbb{R})$ of all infinitely differentiable functions on $\mathbb{R}$ has a Fr'echet space topology with respect to which $f_n \to f$ iff $f_n^{(k)} \to f^{(k)}$ uniformly on compact sets for all $k \geq 0$.

*Solution.* Consider the seminorms $p_{k,n}(f) := \sup_{|x|\leq n} |f^{(k)}|$, consider the TVS generated by $\{p_{k,n}\}_{k,n\geq 0}$ (indexed over integers), this is clearly hausdorff and convergence here is equivalent to uniform convergence of the derivatives on compact sets so we only have to prove completeness. Say $f_n$ is cauchy here, then $f_m^{(k)}$ is cauchy on $[-n,n]$ for any $n$ so we can find $g_n$ defined on $[-n,n]$ so that $f_m^{(k)} \to g_n^{(k)}$ on $[-n,n]$ (this easily follows from Exercise 5.9) If $n \leq m$ then clearly $g_m|_{[-n,n]} = g_n$ so there exists an unique $f$ defined on $\mathbb{R}$ that extends all $g_n$ and for any open ball $B(x,r) \subseteq \mathbb{R}$, for large enough $n$ we have $[-n,n] \supseteq B(x,r)$ and thus $f|_{[-n,n]} = g_n \in C^\infty([-n,n])$ so $f$ is smooth at $x$ and as this was arbitrary, $f \in C^\infty(\mathbb{R})$ proving completeness. □

**Exercise 5.46.** If $X$ is a vector space, $Y$ a normed linear space, $\mathcal{T}$ the weak topology on $X$ generated by a family of linear maps $\{T_\alpha : X \to Y\}$, and $\mathcal{T}'$ the topology defined by the seminorms $\{x \mapsto \|T_\alpha x\|\}$, then $\mathcal{T} = \mathcal{T}'$.

*Solution.* For any net $\langle x_\beta \rangle_{\beta\in B}$ and $\alpha$,

$$T_\alpha(x_\beta) \longrightarrow T_\alpha(x) \iff \| \underbrace{T_\alpha(x_\beta) - T_\alpha(x)}_{=T(x_\beta - x)} \| \longrightarrow 0$$

so convergence of nets is equivalent across these two topologies from Exercise 4.34 and the definitions, making them equal by the lemma in Exercisee 5.43. □

**Exercise 5.47.** Suppose that $X$ and $Y$ are Banach spaces.

a. If $\{T_n\}_1^\infty \subset L(X,Y)$ and $T_n \to T$ weakly (or strongly), then $\sup_n \|T_n\| < \infty$.

b. Every weakly convergent sequence in $X$, and every weak*-convergent sequence in $X^*$, is bounded (with respect to the norm).

*Solution.*

a. If $T_n \to T$ strongly then for all $x, T_n x \to Tx$ so $\sup_n \|T_n x\| < \infty$ given $T_n x$ converges so $\sup_{n\in\mathbb{N}} \|T_n\| < \infty$ by the uniform boundedness principle and if convergence is weak then for all $f \in Y^*$ and $x \in X$, $f(T_n x) \to f(Tx)$ and thus, as $\sup_n \|\widehat{T_n x}(f)\| < \infty$ for all $f \in Y^*$, by the uniform boundedness principle we have $\sup_n \|\widehat{T_n x}\| < \infty \implies \sup_n \|T_n x\| < \infty$ (the natural pairing is an isometry) which implies $\sup_n \|T_n\| < \infty$ after another application of the same.

b. If $x_n \to x$ weakly then $f(x_n) \to f(x)$ for all $f \in X^*$ and equivalently $\widehat{x_n} \to \widehat{x}$ in the strong operator topology over $L(X^*, \mathbb{K})$ implying, by part (a) that $\sup_n \|\widehat{x_n}\| = \sup_n \|x_n\| < \infty$. When $f_n \to f$ is weak*-convergent then for all $x$ we have $\widehat{x}(f_n) = f_n(x) \to f(x) = \widehat{x}(f)$ so we can say that $f_n \to f$ in the strong operator topology in this case and conclude by part (a).

□

**Exercise 5.48.** Suppose that $X$ is a Banach space.

a. The norm-closed unit ball $B = \{x \in X : \|x\| \leq 1\}$ is also weakly closed. (Use Theorem 5.8b., this was "d" in the book)

b. If $E \subset X$ is bounded (with respect to the norm), so is its weak closure.

c. If $F \subset X^*$ is bounded (with respect to the norm), so is its weak* closure.

d. Every weak*-Cauchy sequence in $X^*$ converges. (Use Exercise 38.)

*Solution.*

a. Suppose $\|x_n\| \leq 1$ and $x_n \to x$ weakly, let $f$ be an unit norm linear functional such that $f(x) = \|x\|$ (this exists by theorem 5.8b) then for any $\epsilon > 0$ and large enough $n$ we have $|f(x)| = \|x\| \leq \epsilon + |f(x_n)| \leq \epsilon + \|f\|\|x_n\| \leq 1 + \epsilon$ and as $\epsilon > 0$ was arbitrary we must have $\|x\| \leq 1$ proving that it is weaklt closed.

b. If it is bounded then we can find some $r$ with $E \subseteq \overline{B(x,r)}$ and from the previous part we see that for all sequences $(x_n) \subseteq E \subseteq \overline{B(0,r)}$ which converge, their limit must be in $\overline{B(0,r)}$ too so $\overline{E} \subseteq \overline{B(x,r)}$ proving boundedness.

c. If $F \subseteq \overline{B(0,r)}$ and $(f_n) \subseteq E$ are weak*-convergent then we see that $f_n(x) \to f(x)$ for all $x$ and first of all $f \in X^*$ by Exercise 5.38 and for $\epsilon > 0$ we can find $x$ of unit norm such that $f(x) \geq \|f\| - \epsilon$ and for large enough $n$ we have

$$\|f\| - \epsilon \leq |f(x)| < \epsilon + |f_n(x)| \leq \epsilon + r|x| = \epsilon + r$$

implying $\|f\| \leq r + 2\epsilon$ and as $\epsilon > 0$ was arbitrary we have $\|f\| \leq r$ proving boundedness of the closure just as in part (a).

d. If $(f_n)$ is weak* cauchy then the net $\langle f_n - f_m \rangle_{(n,m)\in\mathbb{N}}$ must converge to 0 in the weak* topologhy and thus for all $x \in X, (f_n(x))$ is a cauchy sequence and as $\mathbb{K}$ is complete, it converges to $\lim f_n(x)$ which we denote by $f(x)$ and $f \in X^*$ by Exercise 5.38 so we are done ($f_n(x) \to f(x)$ follows from construction so we have weak* convergence.)

□

**Exercise 5.49.** Suppose that $X$ is an infinite-dimensional Banach space.

a. Every nonempty weakly open set in $X$, and every nonempty weak*-open set in $X^*$, is unbounded (with respect to the norm).

b. Every bounded subset of $X$ is nowhere dense in the weak topology, and every bounded subset of $X^*$ is nowhere dense in the weak* topology. (Use Exercise 48b,c.)

c. $X$ is meager in itself with respect to the weak topology, and $X^*$ is meager in itself with respect to the weak* topology.

d. The weak* topology on $X^*$ is not defined by any translation-invariant metric. (Use Exercise 48d.)

*Solution.*

a. By the structure of topologies, it suffices to prove this for a base. By Exercise 5.46 and Proposition 5.14 it is sufficient to only consider nonempty sets of form $\cap_1^n U_{x_j f_j \epsilon_j}$ where $U_{xf\epsilon} := \{y : |f(x-y)| < \epsilon\}$. We know from Exercise 5.35 that $f_j \neq 0$ implies $X/\mathcal{N}(f_j)$ is isomorphic to $\mathbb{K}$ and thus $\prod_1^n (X/\mathcal{N}(f_j))$ is isomorphic (atleast algebraically) to $\mathbb{K}^n$ where the prior is a vectorspace under pointwise addition and scalar multiplication. We write $K_j$ for $\mathcal{N}(f_j)$. Define a map $\Phi : X/\cap_1^n K_j \to \prod_1^n (X/K_j)$ by $\pi_j(\Phi(x + \cap_1^n K_j)) := x + K_j$, this is clearly linear and an injection as $\Phi(x + \cap_1^n K_j) = 0$ implies $x + K_j = 0 + K_j$ i.e. $x \in K_j$ for all $j$ and thus $x \in \cap_1^n K_j$ implying $x + \cap_1^n K_j = 0 + \cap_1^n K_j$. If $\{v_\alpha\}_{\alpha\in A}$ is linearly independent in the domain $\Phi$ then by injectivity $\{\Phi(v_\alpha)\}_{\alpha\in A}$ must be lienarly independent in the codomain, which in our case has dimension atmost $n$, being isomorphic to $\mathbb{K}^n$ so the domain can not have dimension more than $n$. Now, if we had $\cap_1^n K_j = \{0\}$ then the domain is $X$ itself but then $X$ would be finite dimensional which is false. So we can find some nonzero $x \in \cap_1^n K_j$ and thus, $y \in \cap_1^n U_{x_j f_j \epsilon_j}$ implies $y + \mathbb{K}x \subseteq \cap_1^n K_j$ (if $f = 0$ then $U_{xf\epsilon} = X$) and the set $y + \mathbb{K}x$ is clearly unbounded given $x \neq 0$ so we are done.
For the second claim, we similarly consider sets of form

$$U_{f\widehat{x}\epsilon} := \{g \in X^* : |\widehat{x}(f-g)| < \epsilon\} = \{g \in X^* : |f(x) - g(x)| < \epsilon\}$$

Say $\cap_1^n U_{f_j \widehat{x_j} \epsilon_j}$ is nonempty and contains $g$. As $X$ is infinite dimensional we have $X \neq \text{Span}\{x_j\}$ so by Proposition 5.8a there exists a nonzero linear functional $f$ that is zero on this span (which is closed by Exercise 5.18) and thus also zero on $x_j$, this implies that $g + \mathbb{K}f \subseteq \cap_n^1 U_{f_j \widehat{x_j} \epsilon_j}$ and that set is clearly unbounded so we are done.

b. The arguement is identical for $X$ with the weak topology and $X^*$ with the weak* topology so we only state it once. By Exercise 5.48b,c (b for the weak case and c for the weak* case) if a set is bounded then so is its closure and by the previous part of this problem, these having nonempty interior would make them unbounded so they must have empty interior, making them nowhere dense.

c. As $X$ and $X^*$ are normed, they can be written as countable unions of bounded sets, for an example: balls of radius diverging to infinity which are centered at 0 and by the previous part we see that all of these are nowhere dense in their repsective topologies so we are done here.

d. If it were metirzable then that would be a complete metric space by Exercise 5.48d which makes $X^*$ non meager by the baire category theorem, contradiction the claim proved in the previous part so it is not even metrizable.

□

**Exercise 5.50.** If $X$ is a separable normed linear space, the weak* topology on the closed unit ball in $X^*$ is second countable and hence metrizable. (But cf. Exercise 49d.)

*Solution.* By Exercise 5.43 we can find a metric $\rho$ (it is even translation invariant) that generates the topology on $X^*$ generated by the seminorms $\{f \mapsto |f(x_n)|\}_{n\in\mathbb{N}}$ where $\{x_n\}$ is a countable dense set in $X$. $\rho$ induces a metric on the closed unit ball and we claim that the topology generated by this metric is the weak* topology on the closed unit ball. If $f_\alpha \to f$ weakly then clearly $f_\alpha(x_n) \to f(x_n)$ for all $n$ implying $\rho(f_n - f) \to 0$ and if $\rho(f_n - f) \to 0$ then $f_\alpha(x_n) \to f(x_n)$ for all $n$ and for any $x \in X$ and $\epsilon > 0$ we can find $n$ such that $\|x_n - x\| < \epsilon$ and as we work in the closed unit ball, $|f_\alpha(x) - f(x)| \leq |f_\alpha(x_n) - f(x_n)| + \|f\|\|x_n - x\| + \|f_\alpha\|\|x_n - x\| < |f_\alpha(x_n) - f(x_n)| + 2\epsilon \to 2\epsilon$ in $\alpha$ and as $\epsilon$ was arbitrary we have $f_\alpha(x) \to f(x)$ for all $x$ implying weak convergence so we can say that these are same topologies by the lemma in Exercise 5.43. Second countablity follows from the fact that this is a compact metric space easily, this standard. □

**Exercise 5.51.** A vector subspace of a normed vector space $X$ is norm-closed iff it is weakly closed. (However, a norm-closed subspace of $X^*$ need not be weak*-closed unless $X$ is reflexive; see Exercise 52d.)

*Solution.* The weak topology is coarser than the norm topology so being weakly closed implies norm-closed. Conversely, if $E$ is norm-closed and $y \notin E$ then we can find $f \in X^*$ such that $f|_E = 0$ but $f(y) \neq 0$ and thus $f^{-1}(\mathbb{K} \setminus \{0\})$ is an open neighbourhood of $y$ disjoint with $E$ as $f$ is continuous in the weak topology by definition so $E^c$ is open in the weak topology implying that $E$ is weakly clsoed. □

**Exercise 5.52.** Let $X$ be a Banach space and let $f_1, \ldots, f_n$ be linearly independent elements of $X^*$.

a. Define $T : X \to \mathbb{C}^n$ by $Tx = (f_1(x), \ldots, f_n(x))$. If $\mathcal{N} = \{x : Tx = 0\}$ and $\mathcal{M}$ is the linear span of $f_1, \ldots, f_n$, then $\mathcal{M} = \mathcal{N}^0$ in the notation of Exercise 23 and hence $\mathcal{M}^*$ is isomorphic to $(X/\mathcal{N})^*$.

b. If $F \in X^{**}$, for any $\epsilon > 0$ there exists $x \in X$ such that $F(f_j) = f_j(x)$ for $j = 1, \ldots, n$ and $\|x\| \leq (1 + \epsilon)\|F\|$. ($F|_{\mathcal{M}}$ can be identified with an element of $(X/\mathcal{N})^{**}$ and hence with an element of $X/\mathcal{N}$ since the latter is finite-dimensional.)

c. If $X$ is considered as a subspace of $X^{**}$, the relative topology on $X$ induced by the weak* topology on $X^{**}$ is the weak topology on $X$.

d. In the weak* topology on $X^{**}$, $X$ is dense in $X^{**}$ and the closed unit ball in $X$ is dense in the closed unit ball in $X^{**}$.

e. $X$ is reflexive iff its closed unit ball is weakly compact.

*Solution.*

a. $\mathcal{M} \subseteq \mathcal{N}^0$ is obvious and by Exercise 5.23, $\mathcal{N}^0$ is isomorphic to $(X/\mathcal{N})^*$ and as $\dim(X/\mathcal{N}) \leq \dim(\prod_j (X/\mathcal{N}(f_j))) = n$ using the arguement in Exercise 5.49 as all $f_j$ are nonzero, and have $\dim((X/\mathcal{N})^*) \leq n$ implying $\dim(\mathcal{N}^0) \leq n$ so $\mathcal{M} = \mathcal{N}^0$ and $\mathcal{M}$ is isomorphic to $(X/\mathcal{N})^*$.

b. $F|_{\mathcal{M}} \in \mathcal{M}^*$ and the map $\Phi : X/\mathcal{N} \to \mathcal{M}^*$ given by $x+\mathcal{N} \mapsto (f \mapsto \widehat{x}(f))$ is injective as $f_j(x) = 0$ for all $j$ implies $x \in \mathcal{N}$ and as $n = \dim(X/\mathcal{N}) = \dim((X/\mathcal{N})^*) = \dim(\mathcal{M}) = \dim(\mathcal{M}^*)$ this map is an isomorphism, thus $F|_{\mathcal{M}} = \Phi(x+\mathcal{N})$ for some $x$ and we can write $F|_{\mathcal{M}}(f_j) = \Phi(x+\mathcal{N})(f_j) = f_j(x)$ so $F(f_j) = f_j(x)$ and $F|_{\mathcal{M}} = \widehat{x}|_{\mathcal{N}^0}$. We claim that $\Phi$ is an isometry, indeed

$$\|\Phi(x+\mathcal{N})\|_{(\mathcal{N}^0)^*} = \sup_{f \in \mathcal{N}^0, \|f\| \leq 1} |f(x)|$$

This is atleast $\|x+\mathcal{N}\|_{X/\mathcal{N}}$ from theorem 5.8a as $\mathcal{N}$ is closed and if $\|f\| \leq 1$ then for all $v \in \mathcal{N}$ we have $|f(x)| = |f(x+v)| \leq \|x+v\|$ as $f(v) = 0$ by definition, thus $|f(x)| \leq \|x+\mathcal{N}\|$ and we thus have equality, proving that this is an isometry. We can now write $\|F|_{\mathcal{M}}\| = \|\Phi(x+\mathcal{N})\| = \|x+\mathcal{N}\|$ and for any $\epsilon > 0$ we can find $v \in \mathcal{N}$ such that $(1+\epsilon)\|x+\mathcal{N}\| \geq \|x+v\|$ (As $\mathcal{N}$ is closed, $\|x+\mathcal{N}\| = 0 \implies x \in \mathcal{N}$) and we still have $F(f_j) = f_j(x+v)$ while $\|F\| \geq \|F|_{\mathcal{M}}\| = \|\Phi(x+\mathcal{N})\| = \|x+\mathcal{N}\| \geq (1+\epsilon)^{-1}\|x+v\|$ so $(1+\epsilon)\|F\| \geq \|x+v\|$.

c. We can use the lemma in Exercise 5.43 and check that the convergence of nets is equivalent in these topologies. $\widehat{x_\alpha} \to \widehat{x}$ in the weak* topology on $X^{**}$ iff $\forall f \in X^*, \widehat{x_\alpha}(f) = f(x_\alpha) \to f(x) = \widehat{x}(f)$ iff $x_\alpha \to x$ in the weak topology on $X$ so we are done.

d. By Exercise 5.46 and proposition 5.14, to show density of $X$ (that is to say, $\widehat{X}$) in $X^{**}$, to prove density at $F \in X^{**}$ it suffices to consider sets of form $\cap_1^n U_{F\widehat{f_j}\epsilon_j}$ (these form a neighborhood base) where $\widehat{f} \in \widehat{X^{**}}$, here $U_{F\widehat{f}\epsilon}$ is the set $\{G \in X^{**} : |\widehat{f}(F-G)| < \epsilon\}$ as in proposition 5.14. By part (b) we can find $x \in X$ such that $F(f_j) = \widehat{x}(f_j)$ and thus $\widehat{x} \in \cap_1^n U_{F\widehat{f}\epsilon}$ proving density of $\widehat{X}$ in $X^{**}$ in the weak* topology and if we consider the closed unit ball $K = \{F \in X^{**} : \|F\| \leq 1\}$ then again from the previous part we can find $\widehat{x} \in \cap_1^n U_{F\widehat{f_j}\epsilon_j}$ such that $\|x\| = \|\widehat{x}\| \leq (1+\epsilon)\|F\| \leq (1+\epsilon)$ with $\epsilon < \epsilon_j(1+\|f_j\|)^{-1}$ for all $j$ and thus $(1+\epsilon)^{-1}\widehat{x} \in K \cap \widehat{X}$, checking the inequalities

$$|(1+\epsilon)^{-1}\widehat{x}(f_j) - F(f_j)| = |\epsilon(1+\epsilon)^{-1}\widehat{x}(f_j)| \leq \epsilon(1+\epsilon)^{-1}\|\widehat{x}\|\|f_j\| < \epsilon_j$$

so we still have $(1+\epsilon)^{-1}\widehat{x} \in \cap_1^n U_{F\widehat{f_j}\epsilon_j} \cap K$ proving the density of $\widehat{X} \cap K$ in $K$.

e. If $X$ is reflexive then $\{F \in X^{**} : \|F\| \leq 1\} = \{\widehat{x} \in \widehat{X} : \|\widehat{x}\| \leq 1\}$ is weak* compact in $X^{**}$ and thus, by part (c) $\{x \in X : \|x\| \leq 1\}$ is weakly compact. Now, if the unit ball in $X$ is weakly compact then the $\{\widehat{x} \in \widehat{X} : \|\widehat{x}\| \leq 1\}$ is weak* compact and weak* dense in the closed unit ball in $X^{**}$ but as this is clearly hausdorff, $\{\widehat{x} \in \widehat{X} : \|\widehat{x}\| \leq 1\}$ is weak* closed and thus all of the closed unit ball in $X^{**}$ which implies reflexivity as we can scale down any $F \in X^{**}$ to be in this unit ball and then scale it back up so we have proven both directions.

□

**Exercise 5.53.** Suppose that $X$ is a Banach space and $\{T_n\}$, $\{S_n\}$ are sequences in $L(X, X)$ such that $T_n \to T$ strongly and $S_n \to S$ strongly.

a. If $\{x_n\} \subset X$ and $\|x_n - x\| \to 0$, then $\|T_n x_n - Tx\| \to 0$. (Use Exercise 47a.)

b. $T_n S_n \to TS$ strongly.

*Solution.*

a. By the mentioned problem in the hint, let $C = \sup_n \|T_n\| < \infty$, then

$$\|T_n x_n - Tx\| \leq \|\|T_n x_n - T_n x\| + \|T_n x - Tx\| \leq C\|x_n - x\| + \|T_n x - Tx\| \to 0$$

b. As $S_n \to S$ strongly, for all $x$ we have $S_n x \to Sx$ and by part (a) we see that $T_n(S_n x) \to T(Sx)$ implying $T_n S_n \to TS$ strongly.

□

**Exercise 5.54.** For any nonempty set $A$, $l^2(A)$ is complete.

*Solution.* We treat $l^2(A)$ as $L^2(A, \mathcal{P}(A), \mu)$ where $\mu$ is the counting measure. For $\{f_n\} \subseteq l^2(A)$, suppose we have $\sum_n \|f_n\| < \infty$, then $|f_n| \leq \|f_n\|$ and we can thus define $f = \sum f_n$ by completeness of $\mathbb{K}$. Let $F_N := \sum_{k \leq N} |f_k|$, then $F_n \to F = \sum |f_k|$ pointwise and by MCT $\int |F|^2 = \lim \int |F_n|^2 = \lim \|F_n\|^2$ and $\|F_n\| \leq \sum_n \|f_n\|$ is bounded by hypothesis so $|F|^2$ is $L^1$. Let $S_n = \sum_{k \leq n} f_n$, then $|f - S_n| \leq F \implies |f - S_n|^2 \leq |F|^2$ so we can use DCT and conclude that,

$$\|f - S_n\|^2 = \int |f - S_n|^2 \longrightarrow 0$$

as $|S_n - f| \to 0$ pointwise, proving completeness (If $\|S_N - f\| < \epsilon$ then $\|f\| \leq \|S_N\| + \|f - S_n\| < \|S_N\| + \epsilon < \infty$ given $S_n \in L^2$ as this has vector space structure, so $f \in L^2$ too.) □

**Exercise 5.55.** Let $\mathcal{H}$ be a Hilbert space.

a. (The polarization identity) For any $x, y \in \mathcal{H}$,

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2).$$

(Completeness is not needed here.)

b. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary iff it is isometric and surjective.

*Solution.*

a. Let $\zeta := \exp(2\pi i/4)$, then four times our sum is

$$\sum_{k=0}^{3} \zeta^k \|x + \zeta^k y\|^2 = \sum_{k=0}^{3} \zeta^k \|x\|^2 + \sum_{k=0}^{3} \zeta^k \Big( \big\langle x, \zeta^k y \big\rangle + \big\langle \zeta^k y, x \big\rangle \Big) + \sum_{k=0}^{3} \zeta^k \|\zeta^k y\|^2$$

The left and right sums are zero as $\sum_0^3 \zeta^k = 0$ expanding the middle sum

$$\sum_{k=0}^{3} \zeta^k \overline{\zeta^k} \, \langle x, y \rangle + \sum_{k=0}^{3} \zeta^k \zeta^k \, \langle y, x \rangle = 4 \, \langle x, y \rangle$$

as $\sum_{k=0}^3 \zeta^{2k} = 0$, this can be checked manually given that $\zeta^{4+j} = \zeta^j$ but a more succint reason for this is the fact that $(k \mapsto n \cdot k) \in \mathrm{Aut}((\mathbb{Z}/m\mathbb{Z})^\times)$ for $n$ coprime to $m$.

b. If $U : \mathcal{H} \to \mathcal{H}'$ is a unitary then it is an isometry as $\|Ux\| = \langle Ux, Ux \rangle = \langle x, x \rangle = \|x\|$ and surjective as it is invertible by definition. Conversely, by the polarisation identity in part (a) and isometry we have

$$\langle Ux, Uy \rangle = \frac{1}{4} \sum_{k=0}^{3} \zeta^k \underbrace{\|Ux + \zeta^k Uy\|^2}_{= \|U(x + \zeta^k y)\|^2} = \frac{1}{4} \sum_{k=0}^{3} \zeta^k \|x + \zeta^k y\| = \langle x, y \rangle$$

so it is an isometry and hence injective, as surjectivity is assumed, it is thus invertible and hence a unitary.

□

**Exercise 5.56.** If $E$ is a subset of a Hilbert space $\mathcal{H}$, $(E^\perp)^\perp$ is the smallest closed subspace of $\mathcal{H}$ containing $E$.

*Solution.* If $F$ is a closed subspace containing $E$ then $E^\perp \supseteq F^\perp \implies (E^\perp)^\perp \subseteq (F^\perp)^\perp$ and as we have $\mathcal{H} = F \oplus F^\perp$ it can be easily shown that $F = (F^\perp)^\perp$ just as we would in the finite dimensional case so $(E^\perp)^\perp \subseteq F$. As $(E^\perp)^\perp$ is itself also a closed subspace the claim is proven. □

**Exercise 5.57.** Suppose that $\mathcal{H}$ is a Hilbert space and $T \in L(\mathcal{H}, \mathcal{H})$.

a. There is a unique $T^* \in L(\mathcal{H}, \mathcal{H})$, called the adjoint of $T$, such that $\langle Tx, y \rangle = \langle x, T^* y \rangle$ for all $x, y \in \mathcal{H}$. (Cf. Exercise 22. We have $T^* = V^{-1} T^\dagger V$ where $V$ is the conjugate-linear isomorphism from $\mathcal{H}$ to $\mathcal{H}^*$ in Theorem 5.25, $(Vy)(x) = \langle x, y \rangle$.)

b. $\|T^*\| = \|T\|$, $\|T^*T\| = \|T\|^2$, $(aS + bT)^* = \overline{a} S^* + \overline{b} T^*$, $(ST)^* = T^* S^*$, and $T^{**} = T$.

c. Let $\mathcal{R}$ and $\mathcal{N}$ denote range and nullspace; then $\mathcal{R}(T)^\perp = \mathcal{N}(T^*)$ and $\mathcal{N}(T)^\perp = \mathcal{R}(T^*)$.

d. $T$ is unitary iff $T$ is invertible and $T^{-1} = T^*$.

*Solution.*

a. As per the hint we can let $V$ be the conjugate linear invertible isometric of $\mathcal{H}$ into its dual given by $y \mapsto f_y$ where $f_y(x) = \langle x, y\rangle$ . Now, $T^\dagger V y$ is the map $x \mapsto f_y \circ T(x) = \langle Tx, y\rangle$ which is also a linear functional, call it $g$, then $g(x) = \langle x, V^{-1}g\rangle$ and thus, $\langle Tx, y\rangle = \left\langle x, V^{-1}T^\dagger V y\right\rangle$. If $S$ was another such map then $\langle Tx, y\rangle = \langle x, Sy\rangle = \langle x, T^*y\rangle$ for all $x, y$ and thus $\langle x, (S - T^*)y\rangle = 0$ so we must have $S_T^* = 0$ by taking $x = (S - T^*)y$ proving uniqueness.

b. All of these identities except the norm ones are proved just as they would be in the finite dimensional case so we assume these and only prove the non trivial ones. As $V$ is an invertible isometry, so is its inverse and thus for all $x \in X$

$$\|T^*x\|_{\mathcal{H}} = \|V^{-1}T^\dagger V x\|_{\mathcal{H}} = \|V^{-1}(T^\dagger V x)\|_{\mathcal{H}} = \|T^\dagger V x\|_{\mathcal{H}^*}$$

and as $\|Vx\|_{\mathcal{H}^*} = \|x\|_{\mathcal{H}}$ and it is also invertible, we have

$$\sup_{\|x\|=1} \|T^*x\|_{\mathcal{H}} = \sup_{\|x\|=1} \|T^\dagger V x\|_{\mathcal{H}^*} = \sup_{\|u\|=1} \|T^\dagger u\|_{\mathcal{H}^*} = \|T^\dagger\|_{L(\mathcal{H},\mathcal{H})} = \|T\|$$

by Exercise 5.22. Now $\|Tx\|^2 = \langle Tx, Tx\rangle = \langle T^*Tx, x\rangle \leq \|T^*Tx\|\|x\| \leq \|T^*T\|\|x\|^2$ by cauchy schwarz so $\|T\|^2 \leq \|T^*T\|$ and $\|T^*T\| \leq \|T^*\|\|T\| = \|T\|^2$ so we are done.

c. $y \in \mathcal{R}(T)^\perp \iff \forall x \in \mathcal{H}, \langle Tx, y\rangle = 0 \iff \forall x \in \mathcal{H}, \langle x, T^*y\rangle = 0 \iff T^*y = 0$ so we have $\mathcal{R}(T)^\perp = \mathcal{N}(T^*)$. The other identity follows by a similar element chase, leveraging the definition of $T^*$.

d. If it is unitary then $\langle Tx, Ty\rangle = \langle x, y\rangle = \langle T^*Tx, y\rangle$ so we must have $T^*T = I$ using the same arguement we used to show uniqueness in part (a) and thus $T^* = T^*TT^{-1} = IT^{-1} = T^{-1}$. Conversely if it is invertible and $T^* = T^{-1}$ then $\langle Tx, Ty\rangle = \langle T^*Tx, y\rangle = \langle x, y\rangle$ as $T^*T = I$ implying that $T$ is unitary by definition.

□

**Exercise 5.58.** Let $\mathcal{M}$ be a closed subspace of the Hilbert space $\mathcal{H}$, and for $x \in \mathcal{H}$ let $Px$ be the element of $\mathcal{M}$ such that $x - Px \in \mathcal{M}^\perp$ as in Theorem 5.24.

a. $P \in L(\mathcal{H}, \mathcal{H})$, and in the notation of Exercise 57 we have $P^* = P$, $P^2 = P$, $\mathcal{R}(P) = \mathcal{M}$ and $\mathcal{N}(P) = \mathcal{M}^\perp$. $P$ is called the orthogonal projection onto $\mathcal{M}$.

b. Conversely, suppose that $P \in L(\mathcal{H}, \mathcal{H})$ satisfies $P^2 = P^* = P$. Then $\mathcal{R}(P)$ is closed and $P$ is the orthogonal projection onto $\mathcal{R}(P)$.

c. If $\{u_\alpha\}$ is an orthonormal basis for $\mathcal{M}$, then $Px = \sum\langle x, u_\alpha\rangle u_\alpha$.

*Solution.*

a. We know that $\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp$ for $x, y \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{K}$ we have $Px, Py \in \mathcal{M}$ and $x - Px, y - Py \in \mathcal{M}^\perp$ so

$$\alpha x + \beta y = \underbrace{\alpha Px + \beta Py}_{\in \mathcal{M}} + \underbrace{\alpha(x - Px) + \beta(y - Py)}_{\in \mathcal{M}^\perp}$$

so $P(\alpha x+\beta y)=\alpha Px+\beta Py$ given that we have a direct sum so all vectors in $\mathcal{H}$ can be written uniquely as the sum of vectors from $\mathcal{M},\mathcal{M}^\perp$. By the pythagorean theorem we have $\|x\|^2=\|Px\|^2+\|x-Px\|^2\geq\|Px\|^2$ so $P\in L(\mathcal{H},\mathcal{H})$. As $Px\in\mathcal{M}$ we have $P(Px)=Px$ by definition as $Px=Px+0$ is the unique representation here ($0\in\mathcal{M}$) implying $P=P^2$. Now, for $x,y\in\mathcal{H}$ we have $\langle Px,y\rangle=\langle Px,Py\rangle+\langle Px,x-Py\rangle=\langle Px,Py\rangle+0=\langle x-(x-Px),Py\rangle=\langle x,Py\rangle-\langle x-Px,Py\rangle=\langle x,Py\rangle+0=\langle x,Py\rangle$ implying $P=P^*$. Clearly $\mathcal{R}(P)=\mathcal{M}$ and by Exercise 5.57d we see that $\mathcal{N}(P)=\mathcal{R}(P^*)^\perp=\mathcal{M}^\perp$ as $P=P^*$.

b. It can be easily shown that $\mathcal{R}(P)=\mathcal{N}(I-P)$ and the later is closed. Also as $P=P^*$ we have $\mathcal{R}(P)^\perp=\mathcal{N}(P^*)=\mathcal{N}(P)$ so indeed $\mathcal{H}=\mathcal{R}(P)\oplus\mathcal{N}(P)$ from which we can deduce that $\langle Px,x-Px\rangle=\langle Px,x\rangle-\langle Px,Px\rangle=\langle Px,x\rangle-\langle P^*Px,x\rangle=\langle Px,x\rangle-\langle Px,x\rangle=0$ as $P^*P=PP=P$ here so it is indeed an orthogonal projection onto $\mathcal{R}(P)$ as $P(x-Px)=Px-PPx=Px-Px=0$ so $Px\in\mathcal{R}(P)$ and $x-Px\in\mathcal{N}(P)=\mathcal{R}(P)^\perp$.

c. Firstly, this basis can be extended to an orthonormal basis of $\mathcal{H}$ using zorn's lemma via arguements like those in proposition 5.28. Say we extend this by attaching a new disjoin set $\{u_\beta\}_{\beta\in B}$ with the indexes being disjoint i.e. $A\cap B=\emptyset$ and then for any $x\in\mathcal{H}$, the set $S=\{\alpha\in A\cup B:\langle x,u_\alpha\rangle\neq 0\}$ is countable. Ennumerating this as $\{s_n\}$ we have, from the continuity of $P$,

$$Px=\lim_{n\to\infty}\sum_{k\leq n}P(\langle x,u_{s_n}\rangle u_{s_n})=\lim_{n\to\infty}\sum_{k\leq n,s_k\in A}\langle x,u_{s_n}\rangle u_{s_n}=\sum_{\alpha\in A}\langle x,u_\alpha\rangle u_\alpha$$

by part (c) of theorem 5.27.

□

**Exercise 5.59.** Every closed convex set $\mathbb{K}$ in a Hilbert space has a unique element of minimal norm. (If $0\in K$ the result is trivial; otherwise, adapt the proof of Theorem 5.24.)

*Solution.* Suppose $\{x_n\}\subseteq K$ is a sequence such that $\|x_n\|\to\inf_{x\in K}\|x\|=$ say, $\delta$. By the paralellogram identity, $\|x_n-x_m\|^2+\|x_n+x_m\|^2=2\|x_n\|^2+2\|y_n\|^2$ and for any $\epsilon>0$ we can find $N$ such that $n\geq N\implies\|x_n\|\leq\delta+\epsilon$, rearranging this

$$\frac{1}{4}\|x_n-x_m\|^2=\frac{\|x_n\|^2+\|x_m\|^2}{2}-\left\|\underbrace{\frac{x_n+y_n}{2}}_{\in K}\right\|^2\leq(\delta+\epsilon)^2-\delta^2=2\epsilon\delta+\epsilon^2$$

which can be made arbitraril small so this sequence is cauchy and hence must converge, let $x=\lim x_n$. As the norm is continuous and $K$ closed we see that $x\in K$ and $\|x\|=\lim\|x_n\|=\delta$. If $y$ is another such element then,

$$\delta^2\leq\left\|\frac{x+y}{2}\right\|^2=\frac{\|x\|^2+\|y\|^2}{2}-\left\|\frac{x-y}{2}\right\|^2\leq\delta^2-\left\|\frac{x-y}{2}\right\|^2\implies\|x-y\|=0$$

implying $x=y$ proving uniqueness. □

**Exercise 5.60.** Let $(X,\mathcal{M},\mu)$ be a measure space. If $E\in\mathcal{M}$, we identify $L^2(E,\mu)$ with the subspace of $L^2(X,\mu)$ consisting of functions that vanish outside $E$. If $\{E_n\}$ is a disjoint

sequence in $\mathcal{M}$ with $X = \bigcup_1^\infty E_n$, then $\{L^2(E_n, \mu)\}$ is a sequence of mutually orthogonal subspaces of $L^2(X, \mu)$ and every $f \in L^2(X, \mu)$ can be written uniquely as $f = \sum_1^\infty f_n$ (the series converging in norm) where $f_n \in L^2(E_n, \mu)$. If $L^2(E_n, \mu)$ is separable for every $n$, so is $L^2(X, \mu)$.

*Solution.* If $f_n \in E_n, f_m \in E_m$ for $n \neq m$ then $\langle f_n, f_m \rangle = 0$ as their product is always zero stemming from the fact that $\{E_n\}$ are mutually disjoint. For $f \in L^2(X)$ we can decompose this as $f = \sum f_n$ with $f_n := \chi_{E_n} f$ which is clearly unique, converges pointwise. As $f_k$ are always nonzero one at a time, $|f - \sum_1^n f_k| \leq \sum_1^\infty |f_k|$ implying $|f - \sum_1^n f_k|^2 \leq |f|^2$ and as $f \in L^2$, $|f|^2$ is integrable so DCT gives us convergence of $|f - \sum_1^n f_k|^2$ to 0 in $L^1$ which implies $\sum_1^n f_k \to f$ in $L^2$ so we are done. Now, if $L(E_n, \mu)$ are all seperable, say $F_n$ is a countable dense set in it, then

$$F := \left\{ \sum_{\alpha \in A} f_\alpha : A \text{ is a finite subset of } \mathbb{N} \text{ and } \forall \alpha \in A, f_\alpha \in F_\alpha \right\}$$

is a countable set and we claim that this is dense in $L^2(X)$. For any $f \in L^2$ and $\epsilon > 0$ we can find $f_n \in F_n$ such that $\|f_n - \chi_{E_n} f\| < \epsilon 2^{-n}$ and by the previous part we can find $N$ such that $\|f - \sum_1^N \chi_{E_n} f\| < \epsilon$ so,

$$\left\| f - \sum_{k=1}^N f_k \right\| \leq \left\| f - \sum_{k=1}^N f \chi_{E_k} \right\| + \sum_{k=1}^N \|f_n - \chi_{E_n} f\| < \epsilon(1 + 1 - 2^N) \leq 2\epsilon$$

proving density of $F$ in $L^2(X)$ making it seperable. □

**Exercise 5.61.** Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be $\sigma$-finite measure spaces such that $L^2(\mu)$ and $L^2(\nu)$ are separable. If $\{f_m\}$ and $\{g_n\}$ are orthonormal bases for $L^2(\mu)$ and $L^2(\nu)$ and $h_{mn}(x, y) = f_m(x) g_n(y)$, then $\{h_{mn}\}$ is an orthonormal basis for $L^2(\mu \times \nu)$.

*Solution.* If $m \neq p$ then by Exercise 2.51b (we can do this because $f_m, f_n \in L^2$ implies that $f_m \overline{f_n} \in L^1$, this is clear from the inequality $|xy| \leq |x|^2 + |y|^2$) we have

$$\begin{aligned} \langle h_{mn}, h_{pq} \rangle &= \int_{X \times Y} f_m(x) g_n(y) \overline{f_p(x) g_q(y)} \, \mathrm{d}(\mu \times \nu)(x, y) \\ &= \Bigg( \underbrace{\int_X f_m(x) \overline{f_p(x)} \, \mathrm{d}\mu(x)}_{= \langle f_m, f_n \rangle_{L^2(X,\mu)} = 0 \text{ as } m \neq n} \Bigg) \left( \int_Y g_n(y) \overline{g_q(y)} \, \mathrm{d}\nu(y) \right) = 0 \end{aligned}$$

So we have orthogonality and similarly we see that

$$\langle h_{mn}, h_{mn} \rangle_{L^2(X \times Y)} = \langle f_m, f_m \rangle_{L^2(X)} \langle g_n, g_n \rangle_{L^2(Y)} = 1 \cdot 1 = 1$$

as $\{f_m\}, \{g_n\}$ were orthonormal basis. This proves that $\{h_{mn}\}$ is an orthonormal set and if $u \in L^2(X \times Y)$ then $u^2 \in L^1(X \times Y)$ so by fubini's thoerem we have (we use the notation in page 67) $u_x^2 \in L^1(Y)$ for a.e. $x$ i.e. $u_x \in L^2(Y)$ for a.e. $x$ so we can write

$$\|u_x\|_{L^2(Y)}^2 = \sum_{n=1}^\infty |\langle u_x, g_n \rangle|^2 \quad \text{for } \mu\text{-a.e. } x$$

by parseval's identity. Now, for fixed $n$, consider the function $x \mapsto \langle u_x, g_n \rangle$ (this is the composition of a continuous function $v \mapsto \langle v, g_n \rangle$ on a measurable function $x \mapsto u_x$, see

proposition 2.34), these are in $L^2(X)$ as $|\langle u_x, g_n\rangle|^2 \leq \|u_x\|^2_{L^2(Y)}$ a.e. and fubini's tells us that this a.e. defined function $x \mapsto \|u_x\|^2_{L^2(Y)}$ is in $L^1(X)$. As $\{f_m\}$ is an orthonormal basis we can write,

$$\|\langle u_x, g_n\rangle\|^2_{L^2(X)} = \sum_{m=1}^{\infty} |\langle\langle u_x, g_n\rangle, f_m\rangle|^2$$

Now, by MCT we have

$$\|u\|^2_{L^2(X\times Y)} = \int \underbrace{\sum_{n=1}^{\infty} |\langle u_x, g_n\rangle|^2}_{=\|u_x\|^2_{L^2(Y)}} \,\mathrm{d}\mu(x) = \sum_{n=1}^{\infty} \|\langle u_x, g_n\rangle\|^2_{L^2(X)} = \sum_{n=1}^{\infty}\sum_{m=1} |\langle\langle u_x, g_n\rangle, f_m\rangle|^2$$

And as $u, h_{mn} \in L^2$ we have $uh_{mn} \in L^1$ so by fubini's

$$\begin{aligned}\langle\langle u_x, g_n\rangle, f_m\rangle = \int \langle u_x, g_n\rangle \overline{f_m}(x)\,\mathrm{d}\mu(x) &= \int \left(\int u_x(y)\overline{g_n(y)}\,\mathrm{d}\nu(y)\right)\overline{f_m(x)}\,\mathrm{d}\mu(x)\\ &= \int u(x,y)\overline{g_n(y)f_m(x)}\,\mathrm{d}(\mu\times\nu)(x,y)\\ &= \langle u, f_{mn}\rangle_{L^2(X\times Y)}\end{aligned}$$

From this we finally conclude that

$$\|u\|_{L^2(X\times Y)} = \sum_{m,n=1}^{\infty} |\langle u, h_{mn}\rangle|^2$$

and we are done by Theorem 5.27b. □

**Exercise 5.62.** In this exercise the measure defining the $L^2$ spaces is Lebesgue measure.

a. $C([0,1])$ is dense in $L^2([0,1])$. (Adapt the proof of Theorem 2.26.)

b. The set of polynomials is dense in $L^2([0,1])$.

c. $L^2([0,1])$ is separable.

d. $L^2(\mathbb{R})$ is separable. (Use Exercise 60.)

e. $L^2(\mathbb{R}^n)$ is separable. (Use Exercise 61.)

*Solution.*

a. As in the proof of Theorem 2.26 consider the same $\phi_n$ and as $|\phi_n| \leq |f|$ we see that $|\phi_n - f|^2 \leq 2(|\phi_n|^2 + |f|^2) \leq 4|f|^2$ and thus by DCT, for any $\epsilon > 0$ we can find $n$ large enough such that $\|\phi_n - f\|^2_{L^2} < \epsilon$. Say $\phi_n = \sum_1^m a_j\chi_{E_j}$ where $E_j$ are disjoint. We also see that $m(E\Delta F) = \|\chi_E - \chi_F\|^2_{L^2}$ and thus by proposition 1.20 we can find a disjoint collections of open intervals $H_j$ with union $U_j$ such that $\|\chi_{U_j} - \chi_{E_j}\|^2_{L^2} < \epsilon$ so,

$$\left\|f - \sum_{j=1}^m a_j\chi_{U_j}\right\|_{L^2} \leq \left\|\sum_{j=1}^m a_j(\chi_{E_j} - \chi_{U_j})\right\|_{L^2} + \left\|f - \sum_{j=1}^m a_j\chi_{E_j}\right\|_{L^2} \leq \sum_{j=1}^m |a_j|\epsilon^{1/2} + \epsilon$$

which can be made arbitrarily small. Now, we can find a finite refinement of $\cup H_j$ made of intervals (not necessarily open) and then get rid of the endpoints from

closed intervals and remove singletons, then rearrange the coeffecients to write $\sum_1^m a_j \chi_{U_j} = \sum_1^k b_k \chi_{I_k}$ a.e. (which *is* equality here) where $I_k$ are all disjoint open intervals. The later can be easily approximated in $L^2$ by continuous functions in the same way we do it in the book too so we are done with the same arguements.

b. As the polynomials are dense in $C([0,1])$ in the uniform norm, they are dense in the $L^2$ norm too and as $C([0,1])$ is dense in $L^2([0,1])$ in the same norm, a two step approximation shows that the polynomials are dense in $L^2([0,1])$ too.

c. The set of all polynomials with rational coeffecients is countable and this is dense in the set of polynomials clearly, which in turn is dense in $L^2([0,1])$ as we saw above, making it seperable.

d. By the mentioned exercise this follows by considering $E_n := (n, n+1]$ for $n \in \mathbb{Z}$ (this is sill countable so changing the indexing set doesn't alter anything) and $L^2([0,1])$ being seperable easily implies that $L^2(I)$ is seperable for any bounded interval $I$ so $L^2((n, n+1])$ are all measurable.

e. A hilbert space is seperable iff it has a countable orthonormal basis by proposition 5.29. As $L^2(\mathbb{R})$ is seperable it has a countable orthonormal basis $\{f_n\}_{n\in\mathbb{N}}$ and as $\mathbb{R}^m$ is $\sigma$-finite for all $m$, using the mentioned exercise we can proceed by induction and see that $\{f_{i_1\dots i_n}\}_{(i_1,\dots,i_k)\in\mathbb{N}^n}$ is am orthonormal basis where $f_{i_1\dots i_n}(x_1,\dots,x_n) := f_{i_1}(x_1)\cdots f_{i_n}(x_n)$, which is countable and this in turn implies that $L^2(\mathbb{R}^n)$ is seperable.

□

**Exercise 5.63.** Let $\mathcal{H}$ be an infinite-dimensional Hilbert space.

a. Every orthonormal sequence in $\mathcal{H}$ converges weakly to 0.

b. The unit sphere $S = \{x : \|x\| = 1\}$ is weakly dense in the unit ball $B = \{x : \|x\| \leq 1\}$. (In fact, every $x \in B$ is the weak limit of a sequence in $S$.)

*Solution.*

a. Say $\{u_n\}_{n\in\mathbb{N}}$ is an orthonormal sequence, as all linear functionals here are of form $u \mapsto \langle u, v\rangle$ for some $v$, bessel's inequality tells us that,

$$\sum_{n=1}^{\infty} |\langle u_n, v\rangle| \leq \|v\| \implies |\langle u_n, v\rangle| \longrightarrow 0 \text{ as } n \to \infty$$

proving weak convergence.

b. We prove the stronger claim of every element in $B$ being a weak limit of some sequence in $S$. Firstly, by part (a) we see that 0 is the weak limit of any orthonormal sequence in $S$, which exist due to $\mathcal{H}$ having infinite dimension. For any $x \in B \setminus \{0\}$, we can find an orthonormal basis $\{u_n\}_{n\in\mathbb{N}}$ where $u_1 = x/\|x\|$ by gram schmidt and let $v_n := (1 - \|x\|)^{1/2} u_n + x \in S$ for $n \geq 2$ so for all $v \in \mathcal{H}$ we have $\langle v_n, v\rangle = (1 - \|x\|)\langle u_n, v\rangle + \langle x, v\rangle \to \langle x, v\rangle$ by part (a).

□

**Exercise 5.64.** Let $\mathcal{H}$ be a separable infinite-dimensional Hilbert space with orthonormal basis $\{u_n\}_1^\infty$.

a. For $k \in \mathbb{N}$, define $L_k \in L(\mathcal{H}, \mathcal{H})$ by $L_k(\sum_1^\infty a_n u_n) = \sum_{k+1}^\infty a_n u_{n-k}$. Then $L_k \to 0$ in the strong operator topology but not in the norm topology.

b. For $k \in \mathbb{N}$, define $R_k \in L(\mathcal{H}, \mathcal{H})$ by $R_k(\sum_1^\infty a_n u_n) = \sum_1^\infty a_n u_{n+k}$. Then $R_k \to 0$ in the weak operator topology but not in the strong operator topology.

c. $R_k L_k \to 0$ in the strong operator topology, but $L_k R_k = I$ for all $k$. (Use Exercise 53b.)

*Solution.*

a. $L_k$ shifts the coeffecients $k$-units to the left. Given any $x \in \mathcal{H}$ we see that $\|L_k x\|^2 = \sum_j |\langle L_k x, u_j\rangle|^2 = \sum_j |\langle x, u_{k+j}\rangle|^2 = \sum_{j>k} |\langle x, u_j\rangle|^2 \longrightarrow 0$ as $\|x\|^2 = \sum_j |\langle x, u_j\rangle|^2 < \infty$ so $L_k \to 0$ strongly but for any $k$ we can have $L_k u_{k+1} = u_1$ so $\|L_k\| \geq 1$ and we can not have convergence in norm.

b. This one is basically a right shift. For any $k$ we have $R_k u_1 = u_{k+1}$ so $R_k$ does not tend to 0 in the strong operator topology but we clearly have $\langle R_k x, u_n\rangle = 0$ for all $k \geq n$ and thus given any $x, u \in \mathcal{H}$ we have,

$$|\langle R_k x, u\rangle| = \left|\sum_{n=1}^\infty \overline{\langle u, u_n\rangle}\langle R_k x, u_n\rangle\right| = \left|\sum_{n>k} \overline{\langle u, u_n\rangle}\langle R_k x, u_n\rangle\right| = \left|\sum_{n=1}^\infty \overline{\langle u, u_{n+k}\rangle}\langle x, u_n\rangle\right|$$

For any $\epsilon > 0$, as the sum $\sum_{n=1}^\infty |\langle x, u_n\rangle|^2$ converges we can choose $N$ such that $\sum_{n=N+1}^\infty |\langle x, u_n\rangle|^2 < \epsilon$ and,

$$|\langle R_k x, u\rangle| \leq \left|\sum_{n=1}^N \overline{\langle u, u_{n+k}\rangle}\langle x, u_n\rangle\right| + \left|\sum_{n=N+1}^\infty \overline{\langle u, u_{n+k}\rangle}\langle x, u_n\rangle\right|$$

By the cauchy schwarz, we can bound the second sum by

$$\left(\sum_{n=N+1}^\infty |\langle u, u_{n+k}\rangle|^2\right)^{1/2}\left(\sum_{n=N+1}^\infty |\langle x, u_n\rangle|^2\right)^{1/2} \leq \|u\|\epsilon^{1/2}$$

For the first part, Exercise 5.63a implies that $\langle u, u_{n+k}\rangle \to 0$ as $k \to \infty$ for each fixed $n$ so these $N$ terms all converge to 0 giving us $\limsup_k |\langle R_k x, u\rangle| \leq \|u\|\epsilon^{1/2}$ which can be made arbitrarily small and thus $\langle R_k x, u\rangle \to 0$ proving conveergrence in weak operator topology.

c. $L_k R_k = I$ follows from the definition and $R_k L_k x = \sum_n \langle x, u_{n+k}\rangle u_n$ implying $\|R_k L_k x\|^2 = \sum_{n>k} |\langle x, u_n\rangle|^2 \to 0$ by parseval's identity so $R_k L_k \to 0$ in the strong operator topology.

□

**Exercise 5.65.** $l^2(A)$ is unitarily isomorphic to $l^2(B)$ iff $\text{card}(A) = \text{card}(B)$.

*Solution.* If $\text{card}(A) = \text{B}$ then there exists a bijection $f : A \to B$ and we define $U(\sum_\alpha a_\alpha u_\alpha) := \sum_\alpha a_\alpha v_{f(\alpha)}$ where $\{u_\alpha\}_{\alpha\in A}$ and $\{v_\beta\}_{\beta\in B}$ are the cannonical basis for $l^2$ spaces mentioned in page 176. Clearly $U$ is linear, surjective given the bijectivity of $f$, and it is an isometry as

$$\|Ux\|^2 = \sum_{\alpha\in A} \left| \left\langle Ux, v_{f(\alpha)} \right\rangle \right|^2 = \sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|x\|^2$$

as $\left\langle Ux, v_{f(\alpha)} \right\rangle = \langle x, u_\alpha\rangle$ (this follows from the definition) and thus it is an isometry and also surjective, making it an unitary isomorphism by Exercise 5.55b. Conversely, if they are unitarily isomorphic via $U$ then for a basis $\{u_\alpha\}_{\alpha\in A}$ we claim that $\{U(u_\alpha)\}_{\alpha\in A}$ is an orthonormal basis for $l^2(B)$. This set is clearly orthonormal and if for some $x \in l^2(B)$ we have $\langle x, U(u_\alpha)\rangle = 0$ for all $\alpha \in A$ then as it is unitary we have $\langle U^{-1}x, u_\alpha\rangle = 0$ too so $U^{-1}x = 0 \implies x = 0$ so by theorem 5.27a this is an orthonormal basis. Now, using the fact from cardinal arithmetic that if $\{K_a\}_{a\in A}$ is a collection of countable sets such that $A$ is infinite then $\cup_{a\in A} K_a$ has cardinality atmost that of $A$ we can show that $\text{card}(B) \leq \text{card}(A)$ using an arguement similar to that in proposition 5.29 which, by symmetry implies equality. □

**Exercise 5.66.** Let $\mathcal{M}$ be a closed subspace of $L^2([0,1], m)$ that is contained in $C([0,1])$.

a. There exists $C > 0$ such that $\|f\|_u \leq C\|f\|_{L^2}$ for all $f \in \mathcal{M}$. (Use the closed graph theorem.)

b. For each $x \in [0,1]$ there exists $g_x \in \mathcal{M}$ such that $f(x) = \langle f, g_x\rangle$ for all $f \in \mathcal{M}$, and $\|g_x\|_{L^2} \leq C$.

c. The dimension of $\mathcal{M}$ is at most $C^2$. (Hint: If $\{f_j\}$ is an orthonormal sequence in $\mathcal{M}$, $\sum |f_j(x)|^2 \leq C^2$ for all $x \in [0,1]$.)

*Solution.*

a. Consider the inclusion linear map $i : \mathcal{M} \to C([0,1])$, if $f_n \to f$ in $L^2$ and $f_n \to g$ in the uniform norm then we see that $\|f - g\|_{L^2} \leq \|f - f_n\|_{L^2} + \|g - f_n\|_{L^2}$ and for any $\epsilon > 0$, for large enough $n$ we have $\|g - f_n\|_u < \epsilon \implies \|g - f_n\|_{L^2} \leq \epsilon$ and $\|f - f_n\|_{L^2} < \epsilon$ adding which we get $\|f - g\|_{L^2} < 2\epsilon$ so $f = g$ in $L^2([0,1])$ and as $L^2([0,1])$ is banach, $\mathcal{M}$ is banach too being a closed subspace and as $C([0,1])$ is banach too here, the closed graph theorem implies that $i$ is continuous so we can find some $C > 0$ with $\|f\|_u \leq C\|f\|_{L^2}$ for $f \in \mathcal{M}$.

b. $f \mapsto f(x)$ is a linear functional on the hilbert space $\mathcal{M}$ (it has the induced inner product and it is complete being a closed subset of a complete space $L^2([0,1])$) and we also have $|f(x)| \leq \|f\|_u \leq C\|f\|_{L^2}$ (thus $\|x \mapsto f(x)\| \leq C$) making it continuous so by theorem 5.25 we can find $g_x \in \mathcal{M}$ such that $f(x) = \langle f, g_x\rangle$ and as $\|f \mapsto f(x)\| = \|g_x\|_{L^2}$ as in the paragraph above theorem 5.25 we see that $\|g_x\|_{L^2} \leq C$.

c. If $\{f_j\}_{j=1}^n$ is an orthonormal sequence in $\mathcal{M}$ then bessel's inequality tells us that $\sum_1^n |f_j(x)|^2 = \sum_1^n |\langle f_j, g_x\rangle|^2 \leq \|g_x\|^2 \leq C^2$ and thus,

$$n = \sum_{j=1}^n \|f_j\|_{L^2}^2 = \sum_{j=1}^n \int_0^1 |f_j(x)|^2 \, \mathrm{d}m(x) \leq C^2$$

so we must have $\dim \mathcal{M} \leq C^2$ as otherwise we could find an orthonormal sequence with $m > C^2$ elements given $m \leq \dim \mathcal{M}$ (the dimension is clearly finite as $n \leq C^2$ always holds) which is false.

□

**Exercise 5.67.** (The Mean Ergodic Theorem). Let $U$ be a unitary operator on the Hilbert space $\mathcal{H}$, $\mathcal{M} = \{x : Ux = x\}$, $P$ the orthogonal projection onto $\mathcal{M}$ (Exercise 58), and $S_n = n^{-1} \sum_0^{n-1} U^j$. Then $S_n \to P$ in the strong operator topology. (If $x \in \mathcal{M}$, then $S_n x = x$; if $x = y - Uy$ for some $y$, then $S_n x \to 0$. By Exercise 57d, $\mathcal{M} = \{x : U^* x = x\}$. Apply Exercise 57c with $T = I - U$.)

*Solution.* If $x \in \mathcal{M}$ then $Ux = x$ so $U^j x = x$ for all $j$ implying $S_n x = n^{-1} \sum_1^{n-1} U^j x = x$. Otherwise, if $x = y - Uy$ for some $y$ then

$$S_n(y - Uy) = \frac{\sum_{j=0}^{n-1} U^j(y - Uy)}{n} = \frac{\sum_{j=0}^{n-1} U^j y - U^{j+1} y}{n} = \frac{(I - U^n)y}{n}$$

and as $U$ is an isometry see that $\|S_n(y - Uy)\| = n^{-1}\|y - U^n y\| \leq n^{-1}(\|y\| + \|U^n y\|) \leq 2n^{-1}\|y\| \to 0$ as $\|U^j x\| = \|x\|$ for all $j$ here, proving that $S_n x \to 0$. Using the notation in the hint, we have $\mathcal{N}(T) = \mathcal{R}(T^*)^\perp$ by the mentioned exercise and as $\mathcal{M} = \mathcal{N}(I - U) = \mathcal{N}(I - U^*)$ ($Ux = x$ iff $x = U^{-1}x$ iff $x = U^* x$ as $U^{-1} = U^*$) we have $\mathcal{M} = \mathcal{N}(I - U^*) = \overline{\mathcal{R}(I - U)}^\perp$. Now, by Exercise 5.56 and Exercise 5.5 we have $\mathcal{M}^\perp = (\overline{\mathcal{R}(I - U)}^\perp)^\perp = \overline{\mathcal{R}(I - U)}$ implying $\mathcal{H} = \mathcal{M} \oplus \overline{\mathcal{R}(I - U)}$ by theorem 5.24. For any $v \in \overline{\mathcal{R}(I - U)}$ and $\epsilon > 0$ we can find $u \in \mathcal{R}(I - U)$ with $\|u - v\| < \epsilon$ and as $\|S_n\| \leq 1$ (this follows from the triangle inequality and the fact that $\|U^j x\| = \|x\|$ for all $j$) we have $\|S_n v\| \leq \|S_n(u - v)\| + \|S_n u\| \leq \epsilon + \|S_n u\|$ where the second term tends to 0 as we saw above, so $\limsup \|S_n v\| \leq \epsilon$ for $\epsilon > 0$ implying convergence to zero. From this we see that if $x = y + z$ with $y \in \mathcal{M}$ and $z \in \mathcal{M}^\top$ then $S_n(y + z) = S_n y + S_n z = y + S_n z \to y = P(y + z)$ as $S_n z \to 0$ as we showed above and we can say that $S_n \to P$ strongly. □